Chapter 1

Theory of elasticity

Ⅳ、Basics of Variational Principles

4.1 Introduction

- The differential form of Elasticity equations are difficult to have exact solutions, because:
  - Complex governing equations
  - Complex boundary conditions
- Approximate methods are necessary. Finite Element Analysis (FEA) is a very important one.
- Approximate solutions are close to the exact ones。
- Previously, when deriving governing equations, we often make several simplified assumptions. This will also lead to the approximation of solutions.
- In another word, there is no absolutely exact solution in the world.

4.1 Introduction

Variational Method

- Variational method is a subset of functional analysis, a type of problems seeking the functional extremum.

- We will discuss the classic variational method, which is a very old branch of mathematical analysis.

- We will not discuss its rigorous mathematical definition here.

- We will only introduce the concept of Variational Method, which will not require too much Math basics..

4.1 Introduction

- **Johann Bernoulli** (1667 - 1748) challenged all the mathematicians in Europe with a problem in 1696, “ Suppose there are two arbitrary points in a vertical plane. When a particle slides down from the higher point to the lower one under the effect of gravity without the consideration of friction, which curve is it supposed to go along to achieve the minimum time cost?

$$Q[y]=\int_{x_A}^{x_B}\sqrt{\frac{1+y'^2}{y}}dx$$

This equation will be derived in detail later on

4.1 Introduction

- This is the famous "**Brachistochrone Problem**" published in the "*Teachers' Journal*" of the same year. The key point is not only to obtain the maximum or the minimum value, but to obtain an unknown function (a curve) to satisfy all the conditions.

- This innovative and interesting problem quickly attracted the most famous mathematicians in the world. The journal published the solution of Hospital (1661-1704)、Jacob Bernoulli (1654-1705)、Leibniz (1646-1716） and Newton (1642-1727） in May 1697.

- Actually they reached the same goal by using different solutions that contain talented ideas. The answer is quite interesting that the "Brachistochrone Problem" is a cycloid. But the framework of the variation method cannot be established only based on these solutions.

4.1 Introduction

- Euler（1707-1783） solved the Brachistochrone Problem and then obtain the general method for solving this kind of problems in 1736. He pointed out in his paper that if the following integral reaches its extremum,

$$Q[y]=\int_{x_1}^{x_2} F[x, y(x), y'(x)]\,dx$$

Then function $y(x)$ must satisfy

$$\frac{\partial F}{\partial y}-\frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right)=0$$

This is the famous Euler Equation.

- Lagrange (1736-1813) improved and simplified Euler's research in the context of mathematical analysis in 1755. And **variational method** was established as a new branch of mathematics.

4.1 Introduction

- This is the classical variational method.
- The main problem can be concluded as selecting a function in a proper function set to make a certain integral achieve its extremum.
- This can be resolved by solving the corresponding Euler equation.
- It seems that the problem is not complex at all, and you only need to solve a differential equation.

  Actually it's not true!
- Almost all the natural laws can be expressed in form of variation principle.
- Variational method can be used to realize mathematic unification. Generally speaking, the definite problem of Mathematical and Physical Equations can be transformed into variation problems.
- Variation method is an approximate method to solve Mathematics and Physics problems. Its basic idea is to transform such problems into the variational one.
- It has promoted a series of approximate methods. Ritz method is a most useful one but its trial function is very hard to select and it's also very difficult to calculate its coefficient matrix.
- With the development of computer, FEM is quickly developed.

4.2 Functional

The research target of variational method is functional, which is the extension of the function concept.

Let's first study an example to understand what is functional.

Suppose y = f(x) is a continuous function in a closed interval and is non-negative.

Then the geometric meaning of the following integral is the **area between the curve and the rectangular coordinate system**.

$$Q[f]=\int_0^1 f(x)dx$$

Apparently for any continuous and non-negative function between [0,1] there will be only one certain corresponding value.

For example when $f(x)=x$

Then $$Q[f]=\int_0^1 xdx=\frac{1}{2}$$

When $f(x)=\sqrt{x}$

Then $$Q[f]=\int_0^1 \sqrt{x}dx=\frac{2}{3}$$

4.2 Functional

- **Function**:

  For two variates x and y, x belongs to a number set D and y values belongs to a number set R.

  If there is a unique y value corresponding a given x value from D, x is called the independent variable and y is its function.

  Alternatively, the value of function y only depends on the independent variable x, which is known as " unique determination".

- In the previous example,

  when the function f(x) is given, Q[f(x)] will be uniquely determined.

  It means we can take variable Q[f(x)] as the function of f(x), which can be also written as Q[f]. f can be regarded as a kind of independent variable, and Q is its function.

  The functional is actually a generalized concept of function.

4.2 Functional

- **So now we can make the following definition:**

  Consider a function set as D. If for any function f(x) from D, there will be only one certain value corresponding to Q. Then variable Q can be called the functional of the function f, denoted as

  $$Q = Q[f(x)] \text{ or } Q = Q[f]$$

  It should be noticed that this is only a very rough definition of the functional concept. But it'll be enough for the following discussion.

- **Attention**

  In a simple term, a functional is the function of a function (not like the composite function).

  A functional is not the same as a function. A function usually depends on the value of its independent variable, while a functional depends on the shape of its independent function.

  For example the functional Q mentioned before changes with function f(x) (different curves in the interval [0,1]). It doesn't depends on the value of a certain x or a certain f, but on the function relationship between x and f in the entire set D. It is the curve that matters.

4.2 Functional

- **Let's study one more example for further understanding.**

Denote the number of zero value of function f(x) in the interval [a,b] as N[f(x)]. Then there will must be a certain N corresponding to an arbitrary real function f(x) in this interval.

For example suppose [a,b] is [0,2π], then

$$N[\cos x] = 2$$

$$N[\sin x] = 3$$

$$N[x^2 - 1] = 1$$

N is the functional of f based on the definition mentioned before.

4.3 Variation

❖ Introduction

- Several concepts related to the functional variation will be introduced in detail. Before we go further,

those concepts related to functions will be extended for better understanding.

For example:

- Function extremum→Functional extremum
- Function continuity→Functional continuity
- Function differential→Function variation
- ……

4.3 Variation

Variation of independent functions

- The increment of the independent variable x of a function y(x) means the difference between its two values. That is

$$\Delta x = x - x_1 \text{。}$$

  Obviously the differential dx of x is also a kind of increment. When it's small enough, we have

$$dx = \Delta x$$

- The increment of the independent function of the functional Q[y(x)] means the difference between its two function values.

$$\Delta y = y(x) - y_1(x)$$

  When the increment is small enough, it is called as variation. It can be denoted as δy (x) or δy. That is

$$\delta y = y(x) - y_1(x)$$

- The independent functions usually have to satisfy certain boundary conditions (sometimes satisfies equations). For example y (a) $= y_a$, y (b) $= y_b$; thus the variation of the independent functions must satisfy the homogeneous boundary condition $\delta y\,|_{x=a} = \delta y|_{x=b} = 0$.
- It should be noticed that $\delta y\,(x)$ is also a function of x, but is very small in the specified interval. Also it is assumed that the independent function y(x) can arbitrarily change in a certain set of functions near $y_1\,(x)$.

4.3 Variation

- **Variation of the independent functions**
- **Permissible functions**

The value range of x should be first set if one wishes to find the extremum of function y(x).

To solve the extremum problem of functionals, one should similarly indicate on what kind of functions the functional depends.

The qualified and optional functions are usually called as permissible function.

- **Closeness of functions**

As mentioned before, the independent function y(x) can arbitrarily change among a certain type of functions near $y_1(x)$.

But how to define that y(x) is very close to $y_1(x)$? Let's look at some examples.

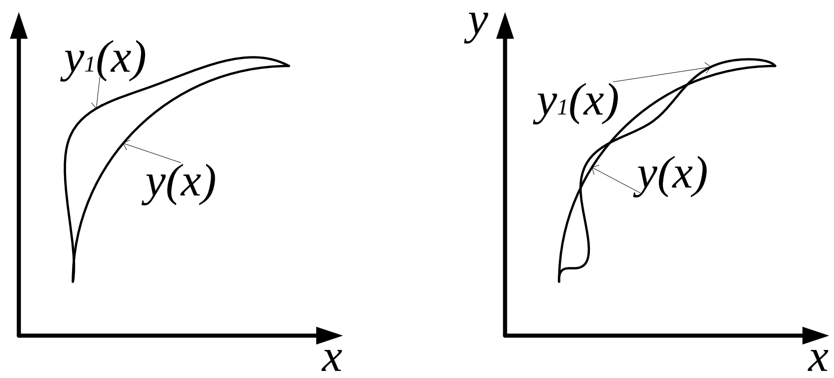
4.3 Variation

The "mode" means $\max|y(x) - y_1(x)|$, $x \in [a, b]$

- **Variation of the independent functio**
  - **Closeness of functions**

Simply speaking, the modes between $y(x)$ and $y_1(x)$ are small for any x, which means the vertical ordinates of curve $y(x)$ and $y_1(x)$ should be very close to each other at any value of abscissa.

The two curves in the following graphs match the conditions obviously but they differ from each other greatly.



In fact:

Not only the vertical ordinates of two curves are similar in the right fig., but their tangential directions too. And the tangential directions of curves in the graph on the left is obviously not close to each other.

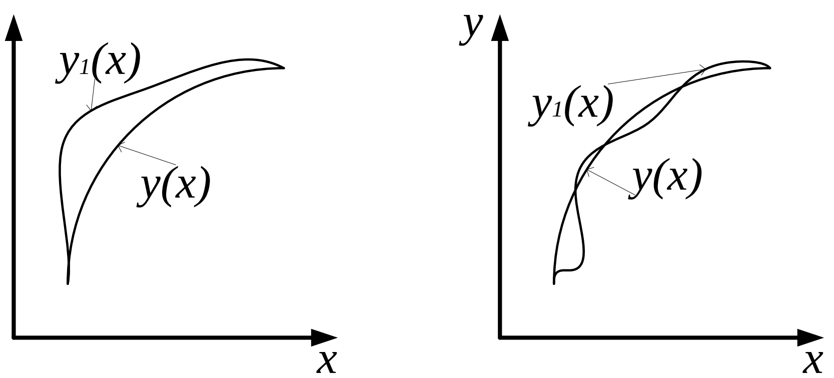
4.3 Variation

❖ Variation of the independent functions

❖ Closeness of functions

The two curves on the left can be called zero-order proximity, of which the difference between y(x) and $y_1(x)$ is rather small at any point, but not for $y'(x)$ and $y'_1(x)$.

The two curves on the right can be called the first-order proximity, of which the difference between y(x) and $y_1(x)$ is small at any point, and the same for $y'(x)$ and $y'_1(x)$.



Sometimes the following differences are required to be very small, which means $\delta y = y(x) - y_1(x)$, $\delta y' = y'(x) - y'_1(x)$, … , $\delta y^{(k)} = y^{(k)}(x) - y^{(k)}{}_1(x)$ are all small.

Then we call the two curves have a K-order proximity. Obviously the higher K is, the better the curves approach to each other.

4.3 Variation

Continuous functional

For an arbitrary small positive number ε, there exists a positive number δ that can make $|\, y(x) - y(x_1)\,| < \varepsilon$ always hold valid when $|x - x_1| < \delta$, then the function is continuous at x_1.

- Definition is the same for the functional.

In a rigorous term, for an arbitrary small positive number ε, there exists a positive number δ that can make $|Q[y(x)] - Q[y(x)]| < \varepsilon$ always hold valid when $|y(x) - y(x1)\,| < \delta$, $|y'(x) - y'(x1)\,| < \delta$, …, $|y^{(k)}(x) - y^{(k)}(x1)\,| < \delta$, then the functional Q[y(x)] is called Kth-order continuous.

4.3 Variation

Continuous functional

- **Example**

For the functional $Q[y(x)] = \int_a^b y(x)dx$

Both y(x) and $y_0(x)$ belong to C[a, b]

if $y(x) \in C[a, b]$, then $Q[y(x)]$ is a continuous functional

Solution

For any $y(x)$, if $\max\limits_{a \le x \le b} |y(x) - y_0(x)| < \dfrac{\varepsilon}{b-a}$

Then $$|Q[y(x)] - Q[y_0(x)]| = \left| \int_a^b [y(x) - y_0(x)]dx \right|$$

$$\le \int_a^b |[y(x) - y_0(x)]| dx < \frac{\varepsilon}{b-a} \int_a^b dx = \varepsilon$$

So it can be proved that $Q[y(x)] = \int_a^b y(x)dx$ is a continuous functional.

4.3 Variation

❖ **Linear functional**

- If the functional $Q[y(x)]$ is linearly, which means

1. $Q[c\,y(x)] = c\,Q[y(x)]$ (c is an arbitrary constant)
2. $Q[y_1(x) + y_2(x)] = Q[y_1(x)] + Q[y_2(x)]$

Then $Q[y(x)]$ is called a linear functional.

These two conditions can be expressed in one equation: $Q[c_1y_1(x) + c_2y_2(x)] = c_1Q[y_1(x)] + c_2Q[y_2(x)]$, which holds valid for any constants c_1 and c_2.

- **For example**

$$Q[y(x)] = \int_a^b y(x)dx$$ is a linear functional

In fact we have

$$Q[c_1y_1(x) + c_2y_2(x)] = \int_a^b [c_1y_1(x) + c_2y_2(x)]dx$$

$$= c_1\int_a^b y_1(x)dx + c_2\int_a^b y_2(x)dx = c_1Q[y_1(x)] + c_2Q[y_2(x)]$$

◆ **However**

$$Q[y(x)] = \int_a^b y^2(x)dx$$ is not a linear one.

You can try to prove it.

4.3 Variation

Differential of functions

Let's first review the differential of functions, which is usually defined in two ways.

- **The conventional definition of differential**

If the increment of the function $\Delta y=y(x+\Delta x)-y(x)$can be expanded into the linear term and the nonlinear term of Δx. That is

$$\Delta y=A(x)\Delta x+\varphi(x,\Delta x)\Delta x^2$$

where $A(x)$is unrelated to Δx but $\varphi(x,\Delta x)$ is related to Δx. And when $\Delta x\to 0$, $\varphi(x,\Delta x)\to 0$. Then $y(x)$ is called differentiable.

- The linear term of the increment is the differential of the function, and we have

$$dy=A(x)\Delta x=A(x)dx=y'(x)dx$$

That means the differential of a function is actually the linear part of its increment.

4.3 Variation

Differential of functions

- The alternative definition of differential

  Suppose that ε is a small parameter, and calculate the derivative of $y(x+\varepsilon\Delta x)$ at ε. That is

$$\frac{\partial}{\partial\varepsilon}y(x+\varepsilon\Delta x)=y'(x+\varepsilon\Delta x)\Delta x$$

  When $\varepsilon\to 0$

$$\frac{\partial}{\partial\varepsilon}y(x+\varepsilon\Delta x)\big|_{\varepsilon=0}=y'(x)\Delta x=dy(x)$$

  That indicates that the derivation of $y(x+\varepsilon\Delta x)$ w.r.t. ε at $\varepsilon = 0$ equals the differential of $y(x)$ at x.

- This definition is similar to that of the variation defined by Lagrange

4.3 Variation

❖ **Variation of functionals**

The functional variation also has two definitions.

For example

The increment of functional $Q[y(x)] = \int_a^b y^2(x)dx$ can be denoted as

$$\Delta Q = Q[y_1(x)] - Q[y(x)] = Q[y(x) + \delta y] - Q[y(x)]$$

$$= \int_a^b [y(x) + \delta y]^2 dx - \int_a^b y^2(x)dx$$

$$= \int_a^b [y^2(x) + 2y(x)\ \delta y + \delta y^2]dx - \int_a^b y^2(x)dx$$

$$= \int_a^b 2y(x)\ \delta y dx + \int_a^b \delta y^2 dx$$

where $\delta y = y_1(x) - y(x)$。

4.3 Variation

You can discuss on your own

❖ Variation of functionals

Thus the increment of this functional is the sum of two terms.

- The first term is

$$\int_a^b 2y(x)\,\delta y dx = T[y(x)\,,\delta y]$$

When function $y(x)$ is fixed, $T[y(x),\delta y]$ is the linear functional of δy.

- The second term is $\int_a^b \delta y^2 dx$

Where $\delta y = y_1(x) - y(x)$, $y(x)$ is a given function and $y_1(x)$ is a arbitrary one. Both $y(x)$ and $y_1(x)$ belong to $C[a,b]$.

If

$$\max_{a\le x\le b} |y_1(x) - y(x)| = \max|\delta y| \to 0$$

Because

$$\left|\int_a^b (\delta y)^2\,dx\right| \le \max_{a\le x\le b}(\delta y)^2\,(b-a)$$

We have

$$\frac{\left|\int_a^b (\delta y)^2\,dx\right|}{\max|\delta y|} \to 0$$

4.3 Variation

Variation of functionals

- Let's carry on the discussion of the second term.

Because

$$\frac{\left|\int_a^b (\delta y)^2\, dx\right|}{\max|\delta y|} \to 0$$

We can deduce that $\int_a^b \delta y^2 dx$ is a higher order infinitesimal than δ*y*, which can be denoted as

$$\int_a^b (\delta y)^2\, dx = o(\delta y)$$

Thus

$$\Delta Q = T[y(x), \delta y] + o(\delta y)$$

- That means the increment of functional $Q[y(x)] = \int_a^b y^2(x)dx$ can be divided into two parts: The first part is a linear functional of δ*y* and the second part is a higher order infinitesimal than δ*y*.
- This conclusion is quite similar to the usual definition of the differential of function. So is T[*y*(*x*),δ*y*] a functional variation?

4.3 Variation

Variation of functionals

- The conventional definition of variation

If y(x) increases by an increment δy in functional Q[y(x)], then the increment of functional Q is

$$\Delta Q = Q[y(x) + \delta y] - Q[y(x)]$$

If ΔQ is denoted as

$$\Delta Q = T[y(x), \delta y] + \beta[y(x), \delta y]$$

Thus T[y(x),δy] is a linear functional for δy (y(x) is given).

And because $\dfrac{\beta[y(x), \delta y]}{\max|\delta y|} \to 0$ (When y(x) is given, max| δy |→0)

Thus T[y(x),δy] can be called the functional variation, written as δQ。

- It can be seen that the variation δQ of the functional Q[y(x)] is the linear part of functional increment ΔQ.

4.3 Variation

❖ Variation of functionals

We introduce another expression of first-order functional variation (no proving for simplification).

- Lagrangan definition of variation

  If the functional variation of $Q[y(x)]$ exist, we fix the value of $y(x)$ and the increment δy. Then

$$Q[y(x)+\alpha\delta y]$$

is an obvious function of variable α, which can be denoted as

$$Q[y(x)+\alpha\delta y]=\varphi(\alpha)$$

And because the derivative $\varphi'(0)$ of function $\varphi(\alpha)$ at $\alpha=0$ equals to the variation of functional $Q[y(x)]$, then

$$\varphi'(0)=\frac{\partial}{\partial\alpha}Q[y(x)+\alpha\delta y]\Big|_{\alpha=0}=\delta Q$$

- So can we propose the similar extremum problem of a functional by taking the analogy of the extreme value of a function ?

4.3 Variation

Functional extremum

- Extreme value of a function

If the values of the function y(x) around the point $x = x_0$ (or called the neighborhood) are smaller (or larger) than $y(x_0)$ at any point, which means

$$dy = y(x) - y(x_0) \leq 0 \quad (\geq 0)$$

Then the function reaches its maximum (or minimum) at the point $x = x_0$.

We also have $$dy = 0$$

- Definition is similar for the functional

- First we introduce the concept "neighborhood of curve $y_0(x)$" for further discussion.

  The set of all curves y(x) that satisfy the following inequality

$$|y(x) - y_0(x)| \leq \varepsilon$$

  is called the ε neighborhood of curve $y_0(x)$.

(more precisely the zero-order ε neighborhood)

4.3 Variation

Functional extremum

- **The strong extremum of functional**

  If any curve y(x) selected in a certain ε neighborhood of curve $y_0(x)$ can always satisfy

  $$\Delta Q = Q[y(x)] - Q[y_0(x)] \leq 0$$

  Then we call the functional $Q[y(x)]$ reaches its strong maximum on the curve $y_0(x)$.

  The strong minimum can be defined similarly.

- However the more commonly used in variation calculus is another concept of "neighborhood" together with the corresponding definition of functional extremum.

If the curve $y(x)$ belongs to the neighborhood of the curve $y_0(x)$, it means not only their values are similar but also their derivatives as well.

1. $|y(x) - y_0(x)| \leq \varepsilon$
2. $|y'(x) - y'_0(x)| \leq \varepsilon$

The set of all the curves $y(x)$ that satisfy the conditions above is called the first-order ε neighborhood of the curve $y_0(x)$.

- The k-order ε neighborhood of the curve $y_0(x)$ can be defined similarly.

4.3 Variation

This integration is actually to calculate the length of curve $y(x)$

❖ **Functional extremum**

Now let's look at a simple question.

- Example

  For the functional

$$Q[y(x)] = \int_0^1 \sqrt{1 + y'^2(x)}dx$$

calculate curve $y_0(x)$ satisfying $y(0) = 0$, $y(1) = 1$ with a continuous first-order derivative that can make Q reaches its minimum.

In terms of the plane geometry we can see that this question is actually about how to calculate the shortest curve passing by the point A (0, 0) and the point B (1, 1).

Obviously $y_0(x)$ is the straight line connecting the points A and B.

But this kind of geometric solution is not satisfying.

Because for an arbitrary functional $Q[y(x)]$, there is generally no obvious geometric meaning.

So how to solve the more generic extremum problems?

4.4 The simplest Euler equations

- If we wish to get the extremum of the functional like

$$Q[y] = \int_a^b F[x, y(x), y'(x)]\,dx$$

we just need to solve the corresponding Euler equations.

$$\frac{\partial F}{\partial y} - \frac{d}{dx}\left(\frac{\partial F}{\partial y'}\right) = 0$$

Now let's first study how to get this equation.

- The differential equations developed earlier than the variational method. So in the early days once the functional extremum problem was turned into the differential equation problems, the problem was done, or at least almost done.
- However after Ritz proposed the approximation method of solving functional extremum problems directly, people found it much more convenient to seek the approximate solution in terms of the variation rather than differential.
- The wide application of computer reinforced this opinion. So people gradually started to aim to transform the differential equation problem into the corresponding functional extremum problem rather than the opposite.
- This chapter will introduce how to transform the functional extremum problem into the boundary problem of differential equations.

4.4 The simplest Euler equations

- **Preliminary theorem of variation**

If the function y = f (x) is continuous in [a, b]

$$\int_a^b f(x)\eta(x)\,dx = 0$$

and the equation above holds valid for any non-zero function η(*x*) that satisfy the following conditions

1. η(*x*) has a continuous derivative in [*a*, *b*]
2. η(*a*) = η(*b*) = 0
3. |η(*x*)|≦ε (ε is an arbitrary give positive number)

Then the function *f* (*x*) identically equals zero in [*a*, *b*].

- We won't prove the conclusion above mathematically, but only give a descriptive proofing.

4.4 The simplest Euler equations

- **Preliminary theorem of variation**

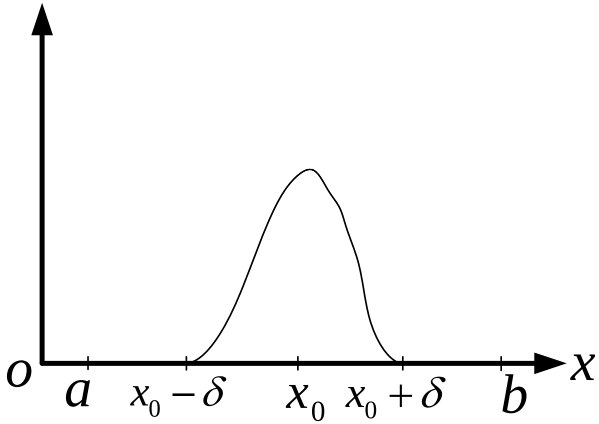
Proving

Using the proof by contradiction we know that if there exists $x_0 \in (a, b)$, that can make $f(x_0)>0$, the conclusion will be contradictory.

Because $f(x)$ is continuous at the point x_0, there must exist a positive number δ that make $f(x)>0$ when $| x - x_0 |<\delta$.

If the function $\varphi(x)$ satisfy

1. when $x \in (x_0 -\delta, x_0 +\delta)$, $\varphi(x)>0$
2. when $x \in [a, x_0 -\delta]$ or $x \in [x_0 +\delta,b]$, $\varphi(x) = 0$
3. $\varphi(x)$ has a continuous derivative in $[a, b]$.



Obviously there will be many functions that satisfy the conditions above, of which the graph on the right is an example.

If $\eta(x) = A\varphi(x)$, then $\eta(x)$ has the three properties above. A is an arbitrary positive constant.

4.4 The simplest Euler equations

- **Preliminary theorem of variation**

For any ε> 0, if there is a proper constant A, we can have

$$|\eta(x)|<\varepsilon$$

For this η(x), integral

$$\int_a^b f(x)\eta(x)\,dx = \int_{x_0-\delta}^{x_0+\delta} f(x)\eta(x)\,dx > 0$$

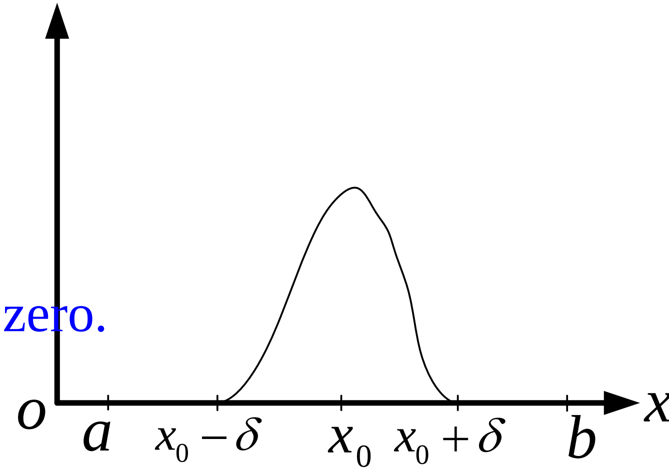
is contradictory to the following equation

$$\int_a^b f(x)\eta(x)\,dx = 0$$

So when $x\in (a, b)$, $f(x)$ is no greater than zero.

And also when $x\in (a, b)$, $f(x)$ is no smaller than zero.

So when $x\in(a, b)$, $f(x)\equiv 0$。



And because $f(x)$ is continuous at the point $x = a$, $x = b$, $f(a) = f(b) = 0$

So when $x\in[a, b]$, $f(x)\equiv 0$

4.4 The simplest Euler equations

Simple Euler equation

- Now let's discuss the extremum conditions of the functional like this

$$Q[y] = \int_a^b F[x, y(x), y'(x)]\,dx$$

- The question can be described as：

  Solve the function $y_0(x)$ in the set D with the following properties, on which the functional Q[$y(x)$] reaches its extremum.

1. $y(x) \in C^1[a,b]$;
2. $y(a) = y_0$, $y(b) = y_1$;
3. *The curve* $y(x)$ is located in a bounded domain B on the plane.

- Now let's review the Fermat theorem of function's extremum in the **differential calculus.**
- Now that the functional is an extended concept of the function, the functional variation is a generalization of function's differential, and the functional extremum is a generalization of function's extremum, will it be possible to extend the Fermat theorem.

4.4 The simplest Euler equations

- **Simple Euler equation**
  - **Theorem**

    Suppose $F(x,y,y')$ is a continuous function of three variables. When the point (x,y) is located in a bounded domain B on the plane with an arbitrary y', $F(x,y,y')$ as well as its first and second order partial derivative is continuous.

    If the functional $Q[y(x)] = \int_a^b F[x, y(x), y'(x)]\,dx$ reaches its extremum on a certain curve y(x) with a continuous second-order derivative when satisfying the following conditions:

1. $y(x) \in C^1[a,b]$;
2. $y(a) = y_0$, $y(b) = y_1$;
3. *The curve y(x) is located in a bounded domain B on the plane.*

Then the function $y(x)$ satisfy the differential equations

$$F_y - \frac{d}{dx} F_{y'} = 0$$

Above is the complete description of Euler theorem.

Next we'll prove the conclusion in detail.

4.4 The simplest Euler equations

- Proving

Now that the functional $Q[y(x)]$ reaches the extremum on the curve $y(x)$. If $y_1(x)$ is a curve in a first-order ε neighborhood of $y(x)$, it satisfies the condition $Q[y_1(x)] \leqq$ （or $\geqq$） $Q[y(x)]$.

Now select a function $\eta(x)$ arbitrarily

1. $\eta(x) \in C^1[a,b]$;
2. $\eta(a) = \eta(b) = 0$;

So when $|\alpha|$ is sufficiently small, the curve $y_1(x) = y(x) + \alpha\eta(x)$ is in the first-order ε neighborhood of $y(x)$.

$y(x)$ and $\eta(x)$ are both given. So the functional $Q[y(x) + \alpha\eta(x)]$ is obviously a function of α. That is $Q[y(x) + \alpha\eta(x)] = \varphi(\alpha)$

The condition of functional extremum discussed above can be equally showed as follows.

When $\alpha = 0$, $\varphi(0) = Q[y(x)]$ reaches its extremum, which means $\varphi(\alpha)$ reaches its extremum at $\alpha = 0$.

According to the Fermat theorem of function extremum it can be deduced that if $\varphi'(0)$ exists, then

$$\varphi'(0) = 0$$

4.4 The simplest Euler equations

- **Prove**

$$\phi'(0)=\phi'(\alpha)\big|_{\alpha=0}=\frac{d}{d\alpha}Q[y(x)+\alpha\,\eta(x)]\Big|_{\alpha=0}$$

$$=\frac{d}{d\alpha}\int_a^b F[x,y(x)+\alpha\,\eta(x),y'(x)+\alpha\,\eta'(x)]dx\Big|_{\alpha=0}$$

Change the order of integral derivation and according to the derivation law of the compound functions we have

$$\phi'(0)=\int_a^b[F_y\frac{d}{d\alpha}(y+\alpha\,\eta)+F_{y'}\frac{d}{d\alpha}(y'+\alpha\,\eta')]dx\Big|_{\alpha=0}$$

$$=\int_a^b[F_y\eta(x)+F_{y'}\,\eta'(x)]dx$$

According to the formula of integration by parts we have

$$\int_a^b[F_y\eta(x)+F_{y'}\,\eta'(x)]dx$$

$$=\int_a^b F_y\eta(x)dx+F_{y'}\,\eta(x)\big|_a^b-\int_a^b\eta(x)\frac{d}{dx}F_{y'}dx$$

- $$=\int_a^b\left(F_y-\frac{d}{dx}F_{y'}\right)\eta(x)dx$$

$\eta(a)=\eta(b)=0$

4.4 The simplest Euler equations

- **Prove**

$$\varphi'(0) = 0$$

Which means

$$\int_a^b \left(F_y - \frac{d}{dx} F_{y'} \right) \eta(x)\, dx = 0$$

Assume $y(x)$ has a second-order continuous derivative, thus

$$\frac{d}{dx} F_{y'} = \frac{d}{dx} F_{y'}(x, y, y') = F_{y'x} + F_{y'y} y' + F_{y'y'} y''$$

So $F_y - \frac{d}{dx} F_{y'}$ is a continuous function.

And function $\eta(x)$ satisfy the following conditions

1. $\eta(x) \in C^1[a,b]$;
2. $\eta(a) = \eta(b) = 0$;

Based on the Preparation theorem of variation we have

$$F_y - \frac{d}{dx} F_{y'} = 0$$

4.4 The simplest Euler equations

So the necessary condition for the functional Q[y(x)] getting extremum is

$$F_y - \frac{d}{dx} F_{y'} = 0$$

which means the function $y(x)$ satisfy the Euler equations.

Combining with the boundary condition, we can find the definite solutions. The functions y(x) that satisfies the Euler equation is called the **stationary zone of the functional Q.** This concept is similar to the station of function.

▪It should be pointed out that the equation above is only a necessary condition but not a sufficient one. So other conditions are needed to confirm if the function $y(x)$ can make the functional get its extremum.

But the so-called " other conditions" will undoubtedly involve more complicated mathematical operations.

▪Fortunately for most of the mathematical problems, the background can provide many intuitive judgement, which means the "other conditions" will not be needed when confirming whether the obtained $y(x)$ can make the functional Q[$y(x)$] get its extremum.

4.4 The simplest Euler equations

- **Example**

Solve the stationary curve of the functional $Q[y(x)] = \int_0^1 (x^2 y + y'^2)dx$

which satisfy the boundary conditions $y(0) = 0$, $y(1) = 1/3$

- Solution

$$F(x, y, y') = x^2 y + y'^2$$

$$F_y = x^2 \quad F_{y'} = 2y'$$

$$\frac{d}{dx} F_{y'} = \frac{d}{dx}(2y') = 2y''$$

Thus the Euler equation is $$x^2 - 2y'' = 0$$

Its general solution is $$y(x) = \frac{1}{24} x^4 + C_1 x + C_2$$

Considering the boundary conditions we can get the stationary zone of the functional

$$y(x) = \frac{1}{24} x^4 + \frac{7}{24} x$$

4.4 The simplest Euler equations

❖ The Brachistochrone problem

➢Physical statement of the problem

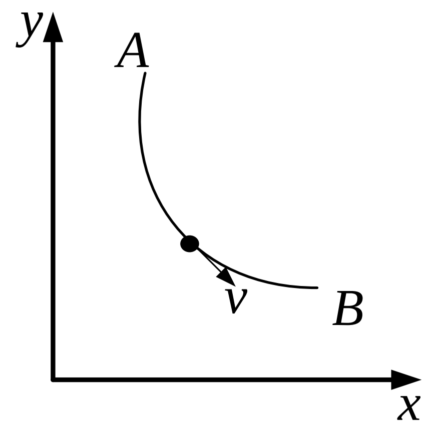
If the plane V is vertical to the ground, and A, B are two arbitrary points on this plane. The point A is higher than B. A particle M descends from A to B along the curve under the effect of gravity and the time it takes is T.

Which curve will make T the minimum?

Of course A is not at the upright top position of B, otherwise it'll be too simple.

Establish a mathematical model

Establish a coordinate system as shown in the graph. Denote the mass of M as m, velocity as v, time as t, then we have



$$\frac{1}{2}mv^2 = mgy \quad \Rightarrow \quad v = \sqrt{2gy}$$

The arc differential of the curve is $dS = \sqrt{1+y'^2(x)}dx$

And

$$dt = \frac{dS}{v} = \frac{\sqrt{1+y'^2}dx}{\sqrt{2gy}}$$

Thus

$$T = \int_{x_A}^{x_B} \sqrt{\frac{1+y'^2}{2gy}}dx = \frac{1}{\sqrt{2g}} \int_{x_A}^{x_B} \sqrt{\frac{1+y'^2}{y}}dx = \frac{1}{\sqrt{2g}} Q[y(x)]$$

4.4 The simplest Euler equations

- **The Brachistochrone problem**
  - Variational expression of the problem

We only need to find a function $y = y(x)$, on which the functional

$$Q[y(x)] = \int_{x_A}^{x_B} \sqrt{\frac{1+y'^2}{y}}dx$$

can attain its extremum.

  - **Establish the Euler equation**

Establish a coordinate system as the graph, Denote the mass of M as m, velocity as v, time as t, then we have

$$F(x, y, y') = \sqrt{\frac{1+y'^2}{y}}$$

Thus

$$F_y = \frac{-\sqrt{(1+y'^2)}}{2y\sqrt{y}} \qquad F_{y'} = \frac{y'}{\sqrt{y(1+y'^2)}}$$

It is obviously hard to carry on with the further calculation of the Euler equation. So let's try another method.

4.4 The simplest Euler equations

- **The Brachistochrone problem**
  - Establish the Euler equation
    Observe that $\mathrm{F}=\mathrm{F}(y,y')$, so

$$F_{y'}=F_{y'}(y,y') \quad \frac{d}{dx}F_{y'}=F_{y'y}y'+F_{y'y'}y''$$

The Euler equation is

$$F_y-\frac{d}{dx}F_{y'}=F_y-F_{y'y}y'-F_{y'y'}y''=0$$

- Now prove

$$F-y'F_{y'}=C \qquad \text{(C is an arbitrary constant)}$$

In fact

$$\frac{d}{dx}[F-y'F_{y'}]=F_yy'+F_{y'}y''-y''F_{y'}-y'(F_{y'y}y'+F_{y'y'}y'')$$

$$=y'(F_y-F_{y'y}y'-F_{y'y'}y'')=0$$

Substitute the expression of F into the equation above, then we have

$$\frac{1}{\sqrt{y(1+y'^2)}}=C\Rightarrow y(1+y'^2)=D$$

(D is also an arbitrary constant)

4.4 The simplest Euler equations

❖ The Brachistochrone problem

- Solving the equation

Let $y' = \tan\theta$, then

$$y = \frac{D}{1+\tan^2\theta} = D\cos^2\theta = \frac{D}{2}(1+\cos 2\theta)$$

thus

$$dy = -D\sin 2\theta\, d\theta$$

$$dx = \frac{dy}{y'} = \frac{-D\sin 2\theta\, d\theta}{\tan\theta} = -D(1+\cos 2\theta)d\theta$$

- So we can get

$$\begin{cases} x = -D\theta - \dfrac{D}{2}\sin 2\theta + E \\ y = \dfrac{D}{2}(1+\cos 2\theta) \end{cases}$$

(D and E are arbitrary constants)

4.4 The simplest Euler equations

The Brachistochrone problem

- Geometric explanation

Let $2\theta = \pi$ - φ, then

$$\begin{cases} x = -D\theta - \dfrac{D}{2}\sin 2\theta + E = \dfrac{D}{2}(\phi - \sin\phi) - D\dfrac{\pi}{2} + E \\ y = \dfrac{D}{2}(1 + \cos 2\theta) = \dfrac{D}{2}(1 - \cos\phi) \end{cases}$$

Suppose when φ = 0, we have x=y=0. Thus

$$\begin{cases} x = r(\emptyset - sin\emptyset) \\ y = r(1 - cos\emptyset) \end{cases} \quad where\ r = \frac{D}{2}$$

- By the Advanced Mathematics we know that the curve above is a cycloid (or called a roulette).
- Exercise: Try to solve the shortest curve connecting the two points using variation method.

4.4 The simplest Euler equations

❖ Variational notations and operational rules

The variational notations and operational rules will be introduced in this chapter for the ease of calculation.

Denote the variation of the functional Q[y(x)] as δQ and the variation of the function y(x) as δy. Let φ(α) = Q[y+ α δy], so

$$\delta Q = \phi'(0) = \int_a^b [F_y \delta y + F_{y'} (\delta y)'] dx$$

It can be proved that

Thus $$(\delta y)' = \delta(y')$$

$$\delta\, y = \underline{y}(x) - y(x)$$
$$(\delta y)' = \underline{y}'(x) - y'(x) = \delta y'$$

$$\delta Q = \int_a^b [F_y \delta y + F_{y'} (\delta y)'] dx = \int_a^b [F_y \delta y + F_{y'} \delta(y')] dx$$

Observe the equation above.

It indicates that calculating the variation δQ of the functional like $\int_a^b F(x, y, y')dx$ is the same as calculating the differential of function because x is fixed and α changes.

4.4 The simplest Euler equations

❖ Variational notations and operational rules

Prove: $$\delta Q = \int_a^b [F_y \delta y + F_{y'} \delta(y')] dx = 0$$

Use the integration by parts to solve the second integral term as before, then

$$\int_a^b F_{y'} \delta(y') dx = \int_a^b F_{y'} d(\delta y) = \delta y F_{y'} \Big|_a^b - \int_a^b \delta y \left(\frac{d}{dx} F_{y'} \right) dx$$

Notice that the variation δy of the independent function equals zero due to the boundary condition, thus we have

$$\delta Q = \int_a^b (F_y - \frac{d}{dx} F_{y'}) \delta y dx = 0$$

As δ*y* is an arbitrary function in the extensive classes of functions, according to the principal theorem of variation we have

$$F_y - \frac{d}{dx} F_{y'} = 0$$

Now let's prove

$$d(\delta y) = \delta(y') dx$$

This kind of transformation is very convenient for variation calculation

It is the interchangeability between differential operator and variation operator.

$$d(\delta y) = (\delta y)' dx = \delta(y') dx = \delta(y' dx) = \delta(dy)$$

4.4 The simplest Euler equations

❖ **Essential and natural boundary conditions**

➢ The boundary conditions mentioned before are **predefined and fixed**.

➢ Thus, the boundary values of the function are known and **irrelative** to the first-order variation of the functional.

➢ This kind of boundary condition is called as the essential boundary condition.

➢ Now let's discuss the variable boundary condition problems. Firstly solve the variation problems as follows

$$\delta Q = \int_a^b (F_y - \frac{d}{dx} F_{y'})\delta y dx + F_{y'}\delta y\Big|_a^b$$

◆ For the first term Euler equation must hold.

◆ For the second term (when x=a and x=b), the following equation must hold.

$$when\ x = a\ and\ x = b \quad we\ have\ \frac{\partial F}{\partial y'} = 0$$

Otherwise a δy can also be found to make δQ not equal zero.

➢ This type of boundary condition is called as the natural boundary condition.

4.4 The simplest Euler equations

- **The second-order variation of functional**

Similar to the **function's** extremum condition, the first-order variation of the functional

$$\delta Q = 0$$

is only the necessary condition for the extremum.

Precisely speaking, it means the functional has the **stationary values** on the function $y(x)$.

The second-order variation also needs to be considered to form the sufficient condition for the functional.

$$\delta^2 Q = \frac{1}{2}\int_a^b \left(\frac{\partial^2 F}{\partial y^2}\delta y^2 + 2\frac{\partial^2 F}{\partial y \partial y'}\delta y \delta y' + \frac{\partial^2 F}{\partial y'^2}\delta y'^2\right)dx$$

must satisfy

$$\delta Q = 0 \; and \; \begin{cases} \delta^2 Q > 0 \;\; minimum \\ \delta^2 Q < 0 \;\; maximum \end{cases}$$

If $\delta^2 Q = 0$, higher order variations have to be taken into consideration.

- We usually consider only the first-order variation in practical application. The sufficiency depends on the physical background of the problem.

So we don't differentiate the stationary value and extremum value in further discussion.

4.4 The simplest Euler equations

- **The conditional extremum of functional (stationary value)**

  The functional extremum problems mentioned before demand no other additional constraint conditions other than the variable functions being smooth and satisfying the given boundary conditions. So they are called the unconditional extremum problems.

  The functional extremum problems with other additional constraint conditions can be called as the conditional extremum problems.

  Usually these additional constraint conditions demand that the independent functions satisfy certain differential or integral relationship in the domain (not only on the boundary).

  The functional extremum problem can be transformed into an equivalent unconditional extremum problem just like the extremum problem of the function using the proper Lagrange multiplier.

  E.g.

  Solve the extremum problem of the functional with definite integral conditions.

4.4 The simplest Euler equations

❖ **The extremum problem of functional with the definite integral condition**

Take an functional w.r.t. an independent function of one variable as example.
Try to obtain an independent function y(x) that can make the functional

$$Q[y(x)] = \int_a^b F(x, y, y')dx$$

reach its extremum in the definition domain $a \leqq x \leqq b$ of independent variable x.
What's more, the function can satisfy the following constraint conditions on its boundary

$$y(a) = y_a; \quad y(b) = y_b;$$

and the integral constraint conditions in the domain

(α is a known constant)
$$\int_a^b \varphi(x, y, y')dx = \alpha$$

Establish a new modified functional by Lagrange multiplier method

$$Q^*[y(x)] = \int_a^b F(x, y, y')dx + \lambda[\int_a^b \varphi(x, y, y')dx - \alpha]$$
$$= \int_a^b F^*(x, y, y', \lambda)dx - \lambda\alpha$$

Where

$$F^*(x, y, y', \lambda) = F(x, y, y') + \lambda\varphi(x, y, y')$$

4.4 The simplest Euler equations

So the problem can be transformed into the unconditional extremum problem of the new functional Q*, where λ is an unknown constant.

The according Euler equation is

$$F_y^* - \frac{d}{dx} F_{y'}^* = 0$$

The general solutions of this equation contain 2 integral constants and a λ, which can be determined by 2 boundary conditions and a definite integral condition. (α is a known constant)

- **Exercise** (**Review the Lagrange multiplier method of the function**)

Obtain the maximum and minimum of z, which is the intersection between the ellipsoid $16x^2 + 4y^2 + z^2 = 16$ and the plane $x + y + z = 1$.

Tips

Considering the unconditional extremums of the function

$$F = z + \lambda_1(x + y + z - 1) + \lambda_2(16x^2 + 4y^2 + z^2 - 16)$$

, where λ1 and λ2 are Lagrange multipliers.

4.5 The more general Euler equations

❖**Summary**

We have derived the Euler equations of the functional like below.

$$Q[y(x)] = \int_a^b F(x, y, y')dx$$

Now we are going to derive the Euler equation of the more general functional using the variational notations and operational rules.

It is not difficult to study the following contents if the previous chapters are well understood. Because all the main difficulties have already been solved, all we have to do is to extend them.

4.5 The more general Euler equations

❖**The functional with the high order derivative of the independent function**

First let's derive the Euler equation of the functional like this

$$Q=\int_a^b F\left(x,y,y',y''\right)dx$$

Suppose the function $F(x, y, y', y'')$ is smooth enough and considering the essential boundary condition.

If the functional gets its extremum on the curve, the variation is

$$\delta Q = 0$$

Derive by the variation notation then we have

$$\delta Q=\int_a^b (F_y\delta y+F_{y'}\delta y'+F_{y''}\delta y'')dx$$

It's known that

$$\int_a^b F_{y'}\delta y'dx=-\int_a^b \delta y\frac{d}{dx}\left(F_{y'}\right)dx$$

4.5 The more general Euler equations

❖**The functional with the high order derivative of the independent function**

Using the integration by parts formula, the third term of the integral can be written as 有

$$\int_a^b F_{y''}\delta y'' dx = \int_a^b F_{y''}\frac{d}{dx}(\delta y')dx = \delta y' F_{y''}\Big|_a^b - \int_a^b \delta y' \frac{d}{dx}\left(F_{y''}\right)dx$$

Because $\delta y'(a) = \delta y'(b) = 0$, we have

$$-\int_a^b \delta y' \frac{d}{dx}\left(F_{y''}\right)dx = -\int_a^b \frac{d}{dx}(\delta y)\frac{d}{dx}\left(F_{y''}\right)dx$$

$$= -\delta y \frac{d}{dx}\left(F_{y''}\right)\Big|_a^b + \int_a^b \delta y \frac{d^2}{dx^2}\left(F_{y''}\right)dx$$

And $\delta y(a) = \delta y(b) = 0$, thus we have

$$\int_a^b F_{y''}\delta y'' dx = \int_a^b \delta y \frac{d^2}{dx^2}\left(F_{y''}\right)dx$$

4.5 The more general Euler equations

❖**The functional with the high order derivative of the independent function**

So the variation of the functional can be denoted as

$$\delta Q = \int_a^b \delta y[F_y - \frac{d}{dx}\left(F_{y'}\right) + \frac{d^2}{dx^2}\left(F_{y''}\right)]dx$$

The integral is smooth.

According to the preliminary theorem of variation, the Euler equation is

$$F_y - \frac{d}{dx}\left(F_{y'}\right) + \frac{d^2}{dx^2}\left(F_{y''}\right) = 0$$

◆This method can be usually used in the similar situation to prove:

for the functional like

$$Q[y(x)] = \int_a^b F(x, y, y', \cdots, y^{(n)})dx$$

the according Euler equation is

$$F_y - \frac{d}{dx}\left(F_{y'}\right) + \frac{d^2}{dx^2}\left(F_{y''}\right) - \cdots + (-1)^n \frac{d^n}{dx^n}\left(F_{y^{(n)}}\right) = 0$$

Exercise: Try to obtain the Euler equation of the extremum problem of the functional containing third-order derivative of its independent function.

4.5 The more general Euler equations

❖ **The functional with several independent functions**

The following statement will be more concise to help understand the above deduction. For the functional like

$$Q[y_1(x),\cdots,y_n(x)]=\int_a^b F(x,y_1,\cdots,y_n,y_1',\cdots,y_n')dx$$

Suppose F $(x, y_1, \ldots, y_n, y_1', \ldots, y_n')$ is smooth enough

If the functional Q reaches its extremum on a set of smooth enough functions that satisfy the given boundary conditions, then $y_i(x)$ will satisfy the Euler equations

$$F_{y_i}-\frac{d}{dx}F_{y_i'}=0 \quad (i=1,2,\cdots,n)$$

4.5 The more general Euler equations

❖ **The functional with multiple independent functions**

Proving

Derive by the variation notations that

$$\delta Q = \int_a^b \left(F_{y_1} \delta y_1 + \cdots F_{y_n} \delta y_n + F_{y_1'} \delta y_1' + \cdots + F_{y_n'} \delta y_n') \right) dx$$

Because

$$\int_a^b F_{y_i'} \delta y_i' dx = \delta y_i F_{y_i'} \Big|_a^b - \int_a^b \delta y_i \frac{d}{dx}\left(F_{y_i'}\right) dx = -\int_a^b \delta y_i \frac{d}{dx}\left(F_{y_i'}\right) dx$$

So

$$\delta Q = \int_a^b [\delta y_1 \left(F_{y_1} - \frac{d}{dx} F_{y_1'} \right) + \cdots + \delta y_n \left(F_{y_n} - \frac{d}{dx} F_{y_n'} \right)] dx = 0$$

δy_i is arbitrarily selected in an extensive class of functions.

4.5 The more general Euler equations

❖ The functional with several independent functions

Prove

Let $\delta y_2 = \ldots = \delta y_n = 0$, the functional will be simplified as

$$\delta Q = \int_a^b \delta y_1 \left(F_{y_1} - \frac{d}{dx} F_{y_1'} \right) dx = 0$$

Because δy_1 is arbitrary and the following function is continuous

$$F_{y_1} - \frac{d}{dx} F_{y_1'}$$

According to the preparation theorem of variation, we have

$$F_{y_1} - \frac{d}{dx} F_{y_1'} = 0$$

The other equations can be proved similarly.

4.5 The more general Euler equations

- **The functional of the function with several variables and the corresponding Euler equations**

We'll give only two examples of the functional containing the functions with several variables. You can try to obtain the general definition on your own.

- The multiple integral

$$Q[z(x,y)] = \iint_D z(x,y)\, dxdy$$

The domain D of the integration is a bounded closed region on the plane xy and $z(x,y)$ is a continuous function of two variables defined in D.

Denote the set of continuous functions of two variables in domain D as C(D). For any $z(x,y) \in$ C(D), the value of Q is uniquely determined. So Q[$z(x,y)$]is the functional w.r.t. $z(x,y)$.

- Supremum of the function

$$Q[z(x,y)] = \sup_G z(x,y)$$

Denote the set of bounded functions in the bounded closed domain G as M(G). For any $z(x,y) \in$M(G)), the value of Q is uniquely determined. So Q[z(x,y)]is the functional based on z(x,y).

4.5 The more general Euler equations

- **The functional of the function with several variables and the corresponding Euler equations**

Now let's discuss the functional

$$Q[z(x,y)] = \iint_D F[x, y, z(x,y), \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}]\, dxdy$$

- Suppose D is a bounded closed domain on the plane *xy*, and the boundary curve ∂D is smooth. The set of the functions with continuous first-order partial derivative is denoted as C^1(D).

- Give a function $\varphi(x,y)$ on the boundary ∂D of the domain D.

And denote the set of the functions z(x, y) in C^1(D) that satisfy the boundary condition

$$z(x,y)\big|_{\partial D} = \phi(x,y)$$

as R^1(D). That is

$$R^1(D) = \left\{ z(x,y) : z(x,y) \in C^1(D),\ z(x,y)\big|_{\partial D} = \phi(x,y) \right\}$$

4.5 The more general Euler equations

- **The functional of the function with several variables and the according Euler equations**

If the functional

$$Q[z(x,y)] = \iint_D F[x, y, z(x,y), \frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}]\, dxdy$$

gets its extremum on the function $z(x,y)$ in $R^1(D)$.

For any $z_1(x,y) \in R^1(D)$ let

$$\delta z = z_1(x,y) - z(x,y)$$

,where δz is the variation of the function $z(x,y)$ and we have

$$\delta z\big|_{\partial D} = [z_1(x,y) - z(x,y)]\big|_{\partial D} = \phi(x,y) - \phi(x,y) = 0$$

That means the variation of the function is zero on the boundary. (The boundary condition here is an essential one)

4.5 The more general Euler equations

❖**The functional of the function with several variables and the according Euler equations**

Let

$$\frac{\partial z}{\partial x} = p, \quad \frac{\partial z}{\partial y} = q$$

Then $$\frac{\partial(\delta z)}{\partial x} = \frac{\partial(z_1 - z)}{\partial x} = \frac{\partial z_1}{\partial x} - \frac{\partial z}{\partial x} = \delta\left(\frac{\partial z}{\partial x}\right) = \delta p$$

Similarly $$\frac{\partial(\delta z)}{\partial y} = \delta q$$

It means the order of differential operation and variation operation can be interchanged.

When $z(x,y)$ and δz are given, the functional $Q[z+\alpha\delta z]$ is obviously the function of α.

Denoted as

$$\phi(\alpha) = Q[z + \alpha\,\delta z]$$

4.5 The more general Euler equations

❖**The functional of the function with several variables and the according Euler equations**

For $$Q[z(x,y)] = \iint_D F(x,y,z,p,q)\,dxdy$$

We have

$$\phi(\alpha) = \iint_D F(x,y,z+\alpha\,\delta z, p+\alpha\,\delta p, q+\alpha\,\delta q)\,dxdy$$

Because the functional Q reaches its extremum on $z(x,y)$, $\phi(\alpha)$ also gets its extremum at $\alpha=0$, which means

$$\phi'(0) = \iint_D (F_z\delta z + F_p\delta p + F_q\delta q)\,dxdy = 0$$

Similarly if the variation of functional $Q[z(x,y)]$ exists, then

$$\delta Q = \phi'(0)$$

Thus

$$\delta Q = \iint_D (F_z\delta z + F_p\delta p + F_q\delta q)\,dxdy = 0$$

4.5 The more general Euler equations

❖**The functional of the function with several variables and the corresponding Euler equations**

However
$$\frac{\partial}{\partial x}\left(F_p\ \delta z\right)=\delta z\frac{\partial}{\partial x}\left(F_p\right)+F_p\frac{\partial}{\partial x}\left(\delta z\right)$$
$$=\delta z\frac{\partial}{\partial x}\left(F_p\right)+F_p\delta\left(\frac{\partial z}{\partial x}\right)=\delta z\frac{\partial}{\partial x}\left(F_p\right)+F_p\delta p$$

So
$$F_p\delta p=\frac{\partial}{\partial x}\left(F_p\ \delta z\right)-\delta z\frac{\partial F_p}{\partial x}$$

Similarly
$$F_q\delta q=\frac{\partial}{\partial y}\left(F_q\ \delta z\right)-\delta z\frac{\partial F_q}{\partial y}$$

Thus $\iint_D(F_p\delta p+F_q\delta q)\ dxdy$
$$=\iint_D[\frac{\partial}{\partial x}\left(F_p\ \delta z\right)+\frac{\partial}{\partial y}\left(F_q\delta z\right)]\ dxdy-\iint_D\delta z(\frac{\partial F_p}{\partial x}+\frac{\partial F_q}{\partial y})\ dxdy$$

4.5 The more general Euler equations

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❖**The functional of the function with several variables and the corresponding Euler equations**

According to the Green formula, we have

$$\iint_D \left[\frac{\partial}{\partial x}\left(F_p\,\delta z\right)+\frac{\partial}{\partial y}\left(F_q\delta z\right)\right] dxdy=\int_{\partial D}\left(F_p\,\delta z l_y - F_q \delta z l_x\right) ds = 0$$

So $$\iint_D (F_p \delta p + F_q \delta q)\, dxdy = -\iint_D \delta z\left(\frac{\partial F_p}{\partial x}+\frac{\partial F_q}{\partial x}\right) dxdy$$

In conclusion the variation of the functional is

$$\delta Q = \iint_D \delta z\left(F_z - \frac{\partial F_p}{\partial x} - \frac{\partial F_q}{\partial y}\right) dxdy$$

So we can solve the according Euler equation due to the preliminary theorem of variation

$$F_z - \frac{\partial F_p}{\partial x} - \frac{\partial F_q}{\partial y} = 0$$

Similarly the solution of this equation is called the polar zone of the functional.

4.5 The more general Euler equations

❖**The functional of the function with several variables and the corresponding Euler equations**

For example

For the functional

$$Q[z(x,y)] = \iint_D \left[\left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2\right] dxdy = \iint_D (p^2 + q^2)\, dxdy$$

Its boundary condition is $z(x,y)\big|_{\partial D} = \phi(x,y)$

What kind of equations z(x, y) should satisfy if the functional Q gets its extremum on this function?

Solution $F(x,y,z,p,q) = p^2 + q^2 \Rightarrow F_p = 2p, F_q = 2q$

Substituted into the Euler equation then we have

$$\begin{cases} \dfrac{\partial^2 z}{\partial x^2} + \dfrac{\partial^2 z}{\partial y^2} = 0 \\ z(x,y)\big|_{\partial D} = \phi(x,y) \end{cases}$$

This is the famous Laplace equation.

4.6 The application of variation method in mechanics

❖Summary

➢The method of variation is widely applied not only in the classical physics and engineering disciplines, but in the modern hi-tech fields such as quantum field theory, control theory and information theory as well.

Take the "brachistochrone problem" for example：

▪Fermat principle

In a homogeneous medium the path taken by a ray of light is the path that can travel in the least time.

▪Einstein general relativity theory

In the four-dimensional Riemann space the path taken by a ray of light is the path that can be travel in the least time.

➢Of course in this chapter we'll mainly introduce the application of variation method in mechanics.

➢ Lagrange established the variation method in the 18th century, which can solve the mechanics problems easily and the principle laws of mechanics as well. That's how the analytical mechanics was established.

4.6 The application of variation method in mechanics

Introduction

- From the history of mechanics, the "*Mathematical principle of natural philos*ophy" written by Newton in 1687 laid the foundation of mechanics by proposing several mechanics laws.

  Newton could perfectly solve the problems of the planets' travelling around the sun with his fundamental laws, which is now thought to be the dynamic problems of free particles or the system of free particles.

- The "Dynamics problems of rigid body" or in a more extensive term, the "Dynamics problems of the constrained particle system", was put forward in the 18th century with the emegence and rapid development of machines.

  One needs to solve the restraint forces first when using the Newton method, which can be very difficult if the machine is relatively complicated.

- The famous work of Lagrange " Analytic mechanics" was published in 1788. The whole mechanics problem can be analyzed by the pure mathematical analysis method based on the variation. What's more important is the proposal of virtual displacement principle, in which the analysis of the restraint forces was not required, as only the external forces appeared in the equilibrium equations.

  The standing of book is sometimes comparable to Newton's "Principles". Because it became the turning point of the history of Mechanics. On one hand it pushed the classical mechanics to its summit, on the other hand it marked the beginning of modern mechanics. The development of Analytical Mechanics is the basis of the modern mechanics especially the quantum mechanics.

  The basic concept of analytical mechanics will be briefly introduced, which will help us understand the application of variation method.

4.6 The application of variation method in mechanics

❖ Hamilton Principle

- When $t = t_1$ and $t = t_2$, the particle is respectively located at A and B. Then its velocity and the real path of its movement should make the integral

$$S = \int_{t_1}^{t_2} (T - U)dt$$

reach its extremum. Where T, U and S are respectively called the kinetic energy , the potential energy, and the action integral.

- The theorem was first proposed by Hamilton in 1834. It is the basic theory of mechanics based on which one can derive the three laws of Newton, *the law of energy conservation, the law of momentum conservation and the law of conservation of the moment of momentum.*

4.6 The application of variation method in mechanics

❖ Hamilton Principle

- Example: Derive the Newton's first law of motion according to the Hamilton Principle.

For a free particle, the kinetic energy is

$$T = \frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

and the potential energy identically equals zero. So the action integral is

$$S = \int_{t_1}^{t_2} \frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)dt$$

When $t = t_1$ the particle is located at the point A(a_1, b_1, c_1); When $t = t_2$ it is located at B (a_2, b_2, c_2) . That is

$$x(t_1) = a_1, y(t_1) = b_1, z(t_1) = c_1$$

$$x(t_2) = a_2, y(t_2) = b_2, z(t_2) = c_2$$

Obviously S[x(t), y(t), z(t)] is the functional of the path。

$$\begin{cases} x = x(t) \\ y = y(t) \\ z = z(t) \end{cases}$$

4.6 The application of variation method in mechanics

Hamilton Principle

- According to the extremum condition of the function with several independent variables mentioned before, we can know that $x(t)$, $y(t)$, $z(t)$ should satisfy the Euler equations.

$$\begin{cases} F_x - \dfrac{d}{dt} F_{\dot{x}} = 0 \\ F_y - \dfrac{d}{dt} F_{\dot{y}} = 0 \\ F_z - \dfrac{d}{dt} F_{\dot{z}} = 0 \end{cases}$$

Because $$F = \frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$$

Then $$F_x = 0, F_{\dot{x}} = m\dot{x} \Rightarrow \frac{d}{dt} F_{\dot{x}} = m\ddot{x} = 0 \Rightarrow \ddot{x} = 0$$

Similarly $$\ddot{y} = 0 \quad \ddot{z} = 0$$

It shows that the acceleration of the free particle equals zero so its velocity is a constant, which means an object either remains at rest or continues to move at a constant velocity, unless acted upon by a force.

4.6 The application of variation method in mechanics

Hamilton Principle

- **Example: Derive the motion equation of the particle in the gravity field.**
- **The acceleration of gravity is supposed to remain the same near the earth surface.**

Suppose the origin of coordinate is located on the earth surface and z-axis goes straight up.

The kinetic energy of the particle is $T = \frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2)$

The potential energy of the particle is $U = mgz$

So the functional

$$S = \int_{t_1}^{t_2} (T - U)dt = \int_{t_1}^{t_2} \left[\frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz \right] dt$$

And

$$F = \frac{m}{2}(\dot{x}^2 + \dot{y}^2 + \dot{z}^2) - mgz$$

Thus we have

$$F_x = F_y = 0 \quad F_z = -mg$$

$$F_{\dot{x}} = m\dot{x} \; F_{\dot{y}} = m\dot{y} \quad F_{\dot{z}} = mg\dot{z}$$

4.6 The application of variation method in mechanics

❖Hamilton Principle

➢Example： **Derive the motion equation of the particle in the gravity field.**

From the according Euler equation we can have

$$\begin{cases} F_x - \dfrac{d}{dt} F_{\dot{x}} = 0 \Rightarrow \ddot{x} = 0 \\ F_y - \dfrac{d}{dt} F_{\dot{y}} = 0 \Rightarrow \ddot{y} = 0 \\ F_z - \dfrac{d}{dt} F_{\dot{z}} = 0 \Rightarrow -mg - m\ddot{z} = 0 \end{cases}$$

So the motion equation of the particle in the gravity field is

$$\begin{cases} \ddot{x} = 0 \\ \ddot{y} = 0 \\ \ddot{z} = -g \end{cases}$$

4.6 The application of variation method in mechanics

❖ Generalized coordinates and the generalized velocity

It is known that there needs to be two coordinates to determine the position of a point on the plane. Actually they don't need to be only the x, y in rectangular coordinate system, but the polar coordinates r, θ as well.

Generally speaking if there are two independent parameters, and

$$\begin{cases} x = x(q_1, q_2) \\ y = y(q_1, q_2) \end{cases}$$

Then q_1 and q_2 are the generalized coordinates of the point on the plane.

- For a particle in rectangular coordinate system space, if there exist independent parameters that

$$\begin{cases} x = x(q_1, q_2, q_3) \\ y = y(q_1, q_2, q_3) \\ z = y(q_1, q_2, q_3) \end{cases}$$

Then q_1, q_2 and q_3 are the generalized coordinates of the particle in space.

- It needs K generalized coordinates $q_1, q_2, \cdots, q_K$ to confirm the position of a system, and hence has K degrees of freedom. And their derivatives to time is

$$\dot{q}_1, \dot{q}_2, \cdots, \dot{q}_K$$

called generalized velocity.

4.6 The application of variation method in mechanics

Lagrange equation

- Suppose the particle has one degree of freedom with one generalized coordinate $q(t)$. And because its position is determined by $q(t)$, the kinetic energy T and the potential energy U are the functions of generalized coordinates and velocity.
- Call T-U the Lagrange function, denoted as

$$T - U = L = L(t, q, \dot{q})$$

Then the action integral of the particle is

$$S = \int_{t_1}^{t_2} L(t, q, \dot{q})dt$$

The real path of the particle make S get its extremum, so

$$\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}} = 0$$

- Similarly for the system with K degrees of freedom

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} = 0 \quad (i = 1, 2, \cdots, K)$$

This is the famous Lagrange equation, the motion equation of the particle in generalized coordinate system that describes the basic laws of motion.

4.6 The application of variation method in mechanics

- **Example**

  Suspend a particle at the end of a filament of the length l. The mass of the particle is m. The filament is fixed at the other end and its mass can be ignored. Write the motion equation of the particle in the gravity field.

  Solve

  Establish the coordinate system as shown in the graph. And the position of this particle is obviously determined by a generalized coordinateθ.

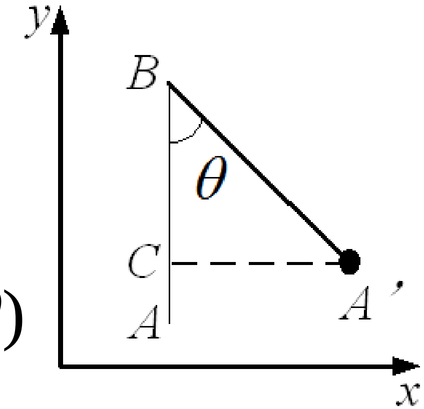
  The kinetic energy $$T=\frac{1}{2}mv^2=\frac{1}{2}m\left(l\dot{\theta}\right)^2$$

  When $\theta=0$, the potential energy is zero, which means the potential energy at θ is

$$U=mg\ AC=mgl(1-\cos\theta)$$

  So the Lagrange function is

$$L=T-U=\frac{1}{2}ml^2\dot{\theta}^2-mgl(1-\cos\theta)$$

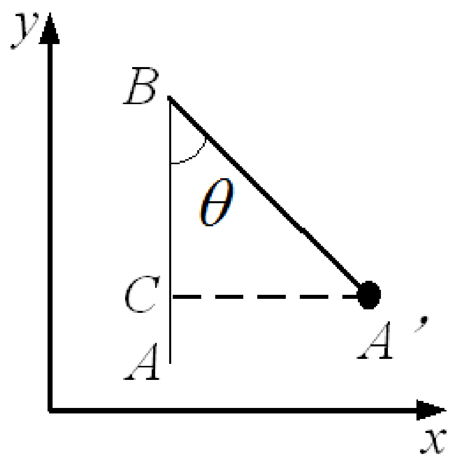


4.6 The application of variation method in mechanics

❖ **Example**

The Lagrange equation of the particle is

$$\frac{\partial L}{\partial \theta} = \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}}$$



And because

$$\frac{\partial L}{\partial \theta} = -mgl\sin\theta \qquad \frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} = \frac{d}{dt}\left(\frac{1}{2}ml^2\dot{\theta}^2\right) = ml^2\ddot{\theta}$$

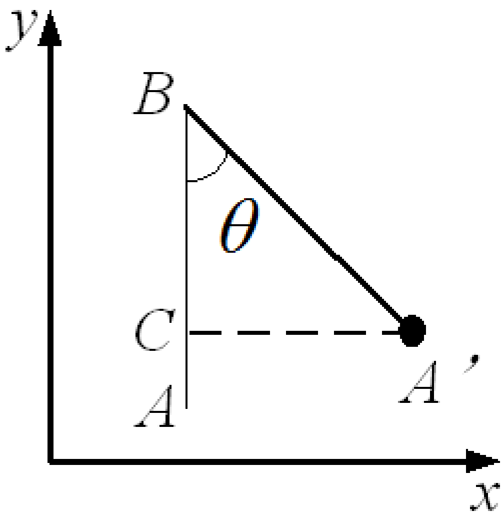
We have

$$\ddot{\theta} + \frac{g}{l}\sin\theta = 0$$

This is the motion equation of a simple pendulum , which is an ordinary second-order nonlinear differential equation.

4.6 The application of variation method in mechanics

- **Example**



If |θ| is small enough, let $\sin\theta \approx \theta$.
Then the equation above can be rewritten as

$$\ddot{\theta} + \frac{g}{l}\theta = 0$$

,which is an ordinary second-order linear differential equation
Its general solution is

$$\theta = c_1 \cos\left(\sqrt{\frac{g}{l}}t\right) + c_2 \sin\left(\sqrt{\frac{g}{l}}t\right)$$

So if the initial condition is

$$\theta(0) = 0 \quad \dot{\theta}(0) = A$$

Then

$$\theta = A\sqrt{\frac{l}{g}} \sin\left(\sqrt{\frac{g}{l}}t\right)$$

The Lagrange equation has a very big advantage in all kinds of coordinate systems. One only needs to write the expressions of the kinetic and potential energy of generalized coordinates and velocity and then substitute them into the Lagrange equation without drawing the force graph.

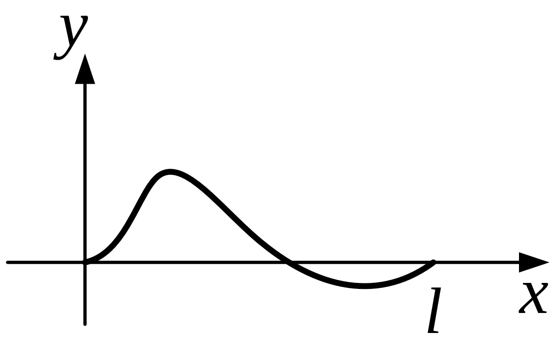
4.6 The application of variation method in mechanics

❖The free transverse vibration of string (both ends fixed)

Establish a coordinate system as shown in the graph. Denote the transverse displacement of point x at time t as the function *u* (*x*,*t*)

Considering a string segment and its length after deformation is

$$dl = \sqrt{1+u_x^2}\,dx$$



The elongation of this segment is

$$\sqrt{1+u_x^2}\,dx - dx = \left(\sqrt{1+u_x^2}-1\right)dx \approx \frac{1}{2}u_x^2\,dx$$

The elastic potential energy is in proportion to its elongation, so we have

$$dU = \frac{K}{2}u_x^2\,dx$$

K is the stiffness of the spring. Thus the potential energy is

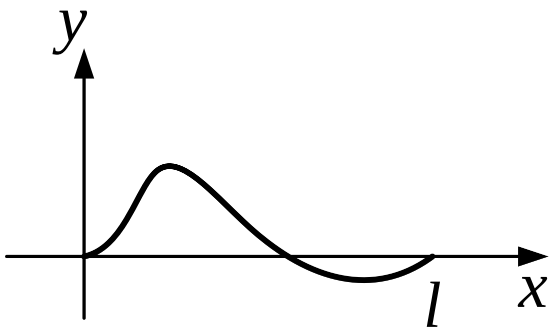
$$U = \int_0^l \frac{K}{2}u_x^2\,dx$$

4.6 The application of variation method in mechanics

❖**The free transverse vibration of string（both ends fixed）**

$\rho(x)$ is the linear density of the string, then the kinetic energy is

$$T=\int_0^l \frac{1}{2}\rho(x)u_t^2\,dx$$



The action integral is

$$S=\int_{t_1}^{t_2}\int_0^l\left(\frac{1}{2}\rho(x)u_t^2(x,t)-\frac{1}{2}Ku_x^2(x,t)\right)dxdt$$

This is the functional of multivariate function $u\,(x,t)$ and its Euler equation is

$$\frac{\partial}{\partial x}\left[Ku_x(x,t)\right]-\frac{\partial}{\partial t}\left[\rho(x)u_t\,(x,t)\right]=0$$

Thus

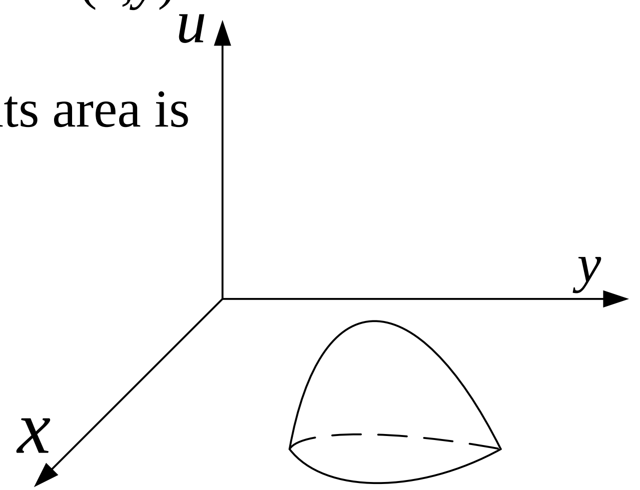
$$u_{tt}(x,t)=\frac{K}{\rho(x)}u_{xx}(x,t)$$

4.6 The application of variation method in mechanics

❖The free transverse vibration of membrane

Establish a coordinate system as shown in the graph.

The function $u\ (x,y,t)$ shows the displacement of the point (x,y) on the membrane at time t.



Take a section of the membrane and the change of its area is

$$\Delta S=\left(\sqrt{1+u_x^2+u_y^2}-1\right)dxdy\approx\frac{1}{2}\left(u_x^2+u_y^2\right)dxdy$$

So its elastic potential energy is

$$dU=\frac{K}{2}\left(u_x^2+u_y^2\right)dxdy$$

Then the general potential energy is

$$U=\iint_\Omega\frac{K}{2}\left(u_x^2+u_y^2\right)dxdy$$

$\rho(x,y)$ is the surface density of the membrane, then the kinetic energy is

$$T=\iint_\Omega\frac{1}{2}\rho(x,y)u_t^2dxdy$$

4.6 The application of variation method in mechanics

❖The free transverse vibration of membrane

Then the action integral is

$$S=\int_{t_1}^{t_2}dt\iint_{\Omega}\left[\frac{1}{2}\rho(x,y)u_t^2-\frac{1}{2}K\left(u_x^2+u_y^2\right)\right]dxdy$$

This is the functional of multivariate function $u(x,y,t)$ and its Euler equation is

$$F_u-\frac{\partial}{\partial x}F_{u_x}-\frac{\partial}{\partial y}F_{u_y}-\frac{\partial}{\partial t}F_{u_t}=0$$

Because

$$F=\frac{1}{2}\rho(x,y)u_t^2-\frac{1}{2}K\left(u_x^2+u_y^2\right)$$

$$F_u=0\quad F_{u_x}=Ku_x\quad F_{u_y}=Ku_y\quad F_{u_t}=\rho u_t$$

So the Euler equation can be written as

$$\frac{\partial}{\partial x}\left(Ku_x\right)+\frac{\partial}{\partial y}\left(Ku_y\right)-\frac{\partial}{\partial t}\left(\rho(x,y)u_t\right)=0$$

Thus

$$u_{tt}=\frac{K}{\rho(x,y)}\left(u_{xx}+u_{yy}\right)$$

4.7 Conclusion

- We can conclude the main steps of variation method from the examples above:
  - **Establish the functional and its boundary conditions based on the physical problems**
  - **Use the preliminary theorem of variation method to obtain the Euler equation by functional variation.**
  - **Solve the Euler equation. This is a differential equation problem.**

- It should be noticed that the variation method and Euler equation describe the same physical problem. Obtaining the proximate solutions from the Euler equation and the variation method have the same effects.

  It's often very difficult to solve the Euler equation, but not that difficult when using the functional variation method. That's why people pay more attention to the latter.
- We have learnt how to obtain the differential equations (Euler equation) of physical equations from the functional variation. But for some problems, the differential equations are known hard to be solved. If they can be transformed into the corresponding extremum problems of functional variation, it will be easily solved by approximation method (eg. finite element analysis).
- However not every differential equation can find the proper functional. In this case we have to go for other kinds of methods to obtain the approximate solutions (eg. the least square method and the Galerkin method), which will be discussed in the following chapters.

V. Elastic Mechanics Problems in the context of Variational Method

5.1 Introduction

- Previously, we mainly discussed the elastic mechanics problems in the context of differential methods.

  - Differential method is to establish and solve a set of basic partial differential equations of elastic mechanics problems under given boundary conditions by the analysis of the action of an arbitrary infinitesimal unit and its adjacent one in an elastic object considering its equilibrium, deformation and material properties.

  - As mentioned before solving the boundary condition problem of partial differential equations is generally very difficult. Once the boundary condition is a little too complicated, it'll be sometimes not impossible to obtain the exact solution in analytical form.

  - Thus people turn to the methods for approximate solutions, among which the variation method stands out due to its advantages.

5.1 Introduction

- **The introduction of the elastic mechanics problems in the variational context**
  - The variational method establishes some functional variation equations considering the energy relationship of the whole elastic system and transform the elastic mechanics problems into solving the variation problems of functional extremum under given constraint conditions.

    Because the functional and the elastic system are both related to energy, the variation principles of elastic mechanics are also called the energy principles and the variation method is called energy method accordingly.

  - So why we still study the variation method when the differential method has already done a good job in analyzing the elastic mechanics problems?

    Actually the differential and variation methods reveal the same physical rule from different aspects, which is the deformation behavior and the final result of a deformable body by the action of external force.

Firstly studying variation method can help understand the elastic mechanics problems.

What’s more, the variation method provides a broader approach to obtain approximate solutions.

5.1 Introduction

- The solution of variation problems
  - The early research work represented by Euler that transforms the variation problems into the corresponding differential equations is called the Euler method.

    This type of research explained the equivalence relation between the variation and differential methods of elastic mechanics. And because the differential equations in the field domain and the boundary conditions can be derived from the unified analysis (functional expressions), such research has a theoretical significance.

    If all the differential equations have solutions, then the exact solutions of the variation problems can be solved.

    Only for very few problems

  - Ritz, Galerkin et all proposed several approximate methods of solving the variation problems directly, which provide a new approach to solving variation problems called the direct methods.

    Fast and efficient calculating methods by computer and commercial programs have been developed on the base of direct methods (mostly the finite element).

    So it promoted the trend of transforming differential equations into the corresponding variation problems.

    But not every differential equation can have its functional form, which means not every physical system has its functional. One can take the non-variational method such like the weighted residual method to solve this kind of problems.

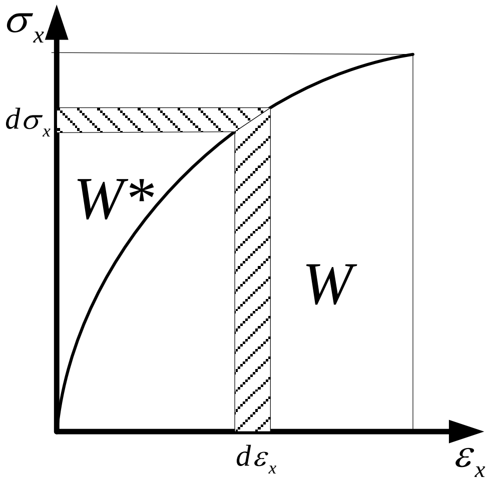
5.2 The strain and complementary energies

❖ Strain energy

As explained before, there exists the strain energy function in an ideal elastic body of small deformation by the action of external force.

Define the strain energy per unit volume as the strain energy density (strain energy for short), denoted as W

$$W = \int_0^{\varepsilon_{ij}} \sigma_{kl} d\varepsilon_{kl}$$



Take the stress condition in one dimension as example.

The graph on the right describes the stress-strain curve of uniaxial tension of an ideal elastic body

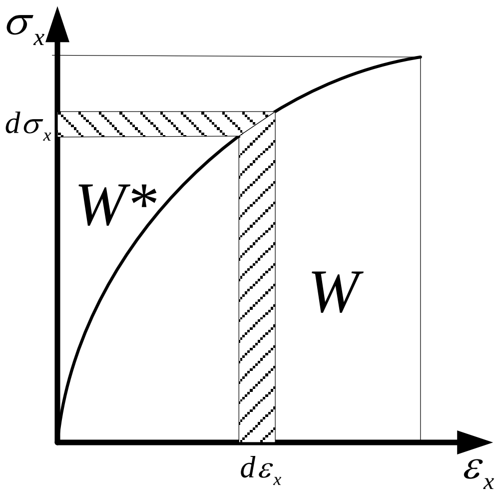
So the geometrical meaning of the integral $W = \int_0^{\varepsilon_x} \sigma_x d\varepsilon_x$ is the area bounded by the stress-strain curve and the horizontal axis.

5.2 The strain and complementary energies

- **Complementary energy**

  There is another important scalar function W*, which is

$$W^* = \int_0^{\sigma_{ij}} \varepsilon_{kl} d\sigma_{kl}$$



  called the complementary energy per unit volume.

  The geometrical meaning of this integral

$$W = \int_0^{\sigma_x} \varepsilon_x d\sigma_x$$

  in an uniaxial tension test is the area closed by the stress-strain curve and the vertical coordinate.

- Apparently we have

$$W + W^* = \sigma_{kl}\varepsilon_{kl}$$

The strain and complementary energies are complementary.

5.2 The strain and complementary energies

❖ **Brief explanation**

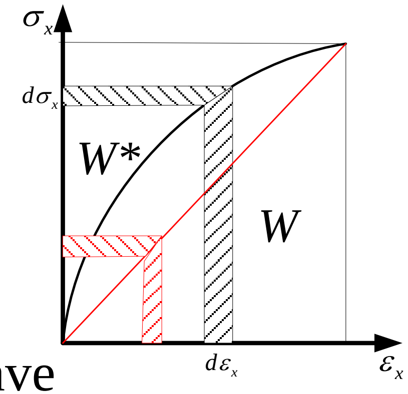
There should be a linear relationship between stress and strain for the linearly elastic body. So we have

$$W = W^* = \frac{1}{2}\sigma_{kl}\varepsilon_{kl}$$

The equation shows that the strain energy of the scalar function equals to its complementary energy in elastic condition. However sometimes we need to differentiate them for theoretical analysis.

We usually express strain energy by a function of strain according to the constitutive relation and express the complementary energy by a function of stress.

$$W = \frac{1}{2}c_{ijkl}\varepsilon_{ij}\varepsilon_{kl} \qquad W^* = \frac{1}{2}d_{ijkl}\sigma_{ij}\sigma_{kl}$$



❖ The Green theorem as mentioned $\sigma_{ij} = \dfrac{\partial W}{\partial \varepsilon_{ij}}$

❖ The complementary energy expressed by stress components is also a single-valued state function and we have

$$\varepsilon_{ij} = \frac{\partial W^*}{\partial \sigma_{ij}}$$

5.2 The strain and complementary energies

- **Brief explanations**

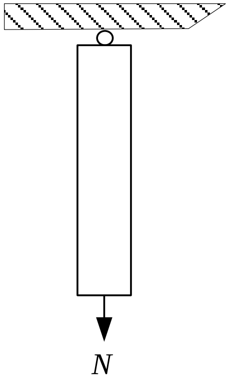
It should be noticed that both the strain and complementary energies are the state function of the ideal elastic body under the action of external force and irrelevant to the loading process.

And the expressions above hold only when there is no initial stress and no deformation in the original state.

Even though the stress-strain relationship of an ideal elastic body approximates to a straight line when the deformation is small enough, the strain and the complementary energies are different although may be the same numerically.

- **Let's think about this simple question for a better understanding**

The tie bar is hinged at one end and experience an axial tension N at the other end. We can obtain the N (tension)-Δ (elongation) curve through test, which can be either linear or non-linear.



5.2 The strain and complementary energies

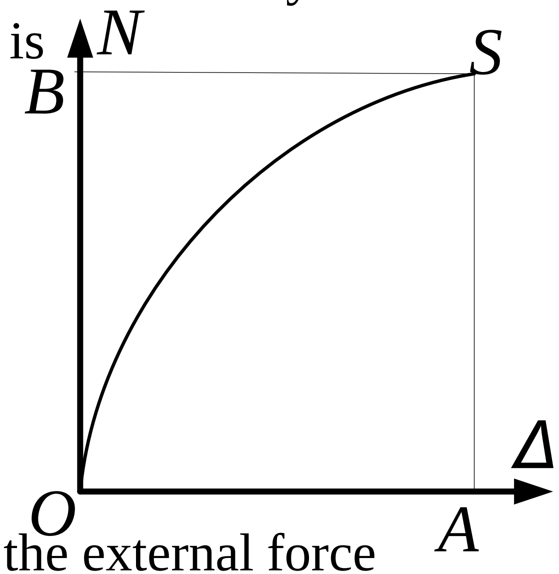
- **Brief explanation**

Suppose the test curve is as shown in the figure.

The elongation is Δ when applied an external force N.

The elongation increases by $d\Delta$ when the external force increases by dN.

The work dU of the external force during this process is



$$dU \approx \left(N + \frac{1}{2} dN \right) d\Delta \approx N d\Delta$$

Thus the work of the external force during the whole loading process is

$$U = \int dU = \int_0^{\Delta} N d\Delta$$

According to the conservation of energy, the work of the external force equals to the strain energy stored in the elastic body. U is the strain energy and its geometrical meaning is the area of the curved triangle OSA.

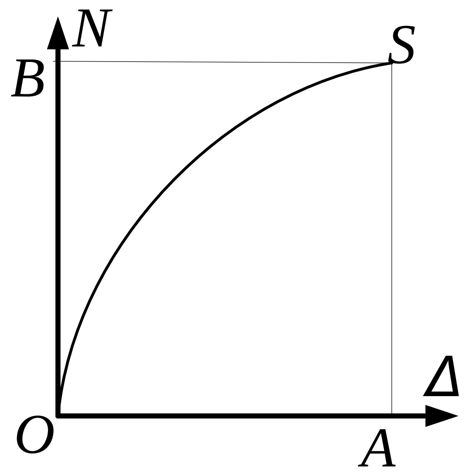
We can get the relation between the tension and the elongation, $N = \dfrac{dU}{d\Delta}$

where U is regarded as the function of Δ.

5.2 The strain and complementary energies

❖ **Brief explanation**

The complementary U* of the bar is defined as the area of the curved triangle OBS.



$$U^{*}=\int_{0}^{N}\Delta dN=N\Delta-U$$

If U* is regarded as the function of N, then we have

$$\Delta=\frac{dU^{*}}{dN}$$

Consider loading the bar with a weight W for understanding of the physical meaning of complementary energy.

The weight cannot be suddenly applied to the bar so it needs a loading device (for example held by hands) for slow loading to maintain the balance.

Suppose N′ is the force applied to the loading device. According to the equilibrium condition we have

$$N+N'=W$$

5.2 The strain and complementary energies

- **Brief explanation**

The curves N-Δ and N′-Δ are shown in the picture on the right.

N and N′ are drawn in the opposite directions for easier observation

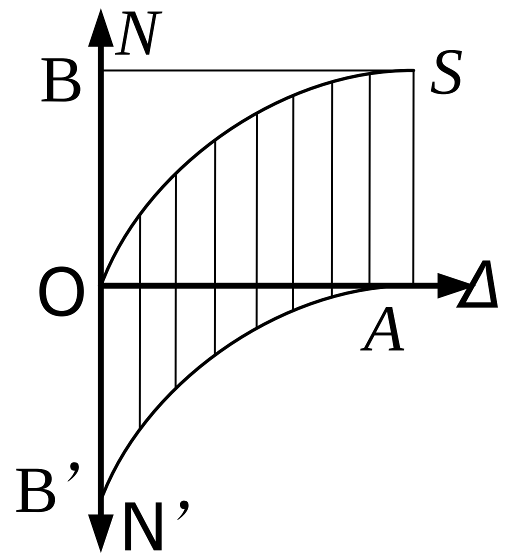
Because N+N′=W

The length of each vertical line parallel to NN' equals W.

Thus

The area of the curved quadrilateral OB'AS equals to the rectangular OASB.

The area of the triangle OB'A equals to the triangle OBS.



The area of the triangle OB'A suggests the work of force applied to the loading device by the weight.

Thus we can deduce that the complementary energy is the work of the loading device under external force.

5.2 The strain and complementary energies

- **The elastic system**

The elastic system consists of the elastic body, the loading system and the support system.

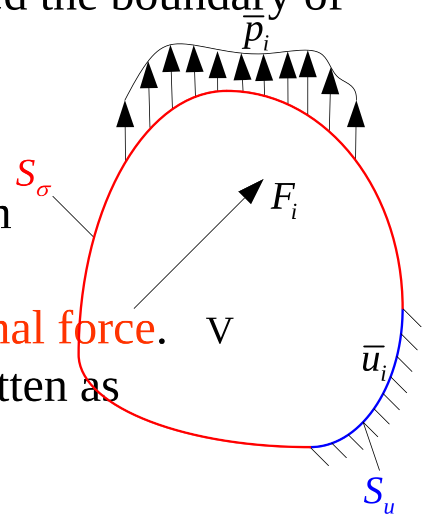
The loading system includes the body force F_i in domain V and the surface force $\overline{p}_i$ on the boundary S_σ of the external force. The letter with a bar above means the given quantity is already known.

The support system only deal with the condition on the certain displacement boundary, which means the displacement value $\overline{u}_i$ is given on S_u, which is called the boundary of fixed supported when the displacement equals zero.

- **The total potential energy Π of the elastic system**

Π is defined as the sum of the strain energy U of the elastic body and the potential energy V of the external force in the loading system

Π = U + V



Now we'll introduce the concept of the potential energy of the external force.

The work A applied on the displacement **u** by the force **F** can be written as

$$A = \int_0^{\bar{\mathbf{u}}} \mathbf{F} \cdot d\mathbf{u} = \int_0^{\overline{u}_i} F_i du_i$$

If the work a force system only depends on the final value of the displacement but not the path, it can be called the conservative force system or the non-conservative force system otherwise.

The integrand function should be total differential to ensure that the work is irrelevant to the path. $F_i du_i = -dV(u_i)$

5.2 The strain and complementary energies

- **The total potential energy Π of the elastic system**

Then

$$A = \int_0^{\bar{u}_i} F_i du_i = -\int_0^{\bar{u}_i} dV(u_i) = -V(\bar{u}_i)$$

So V is also a continuous single-valued state function irrelevant to the path, which is called the potential energy of the force system.

The potential energy is the work capability of the force system.

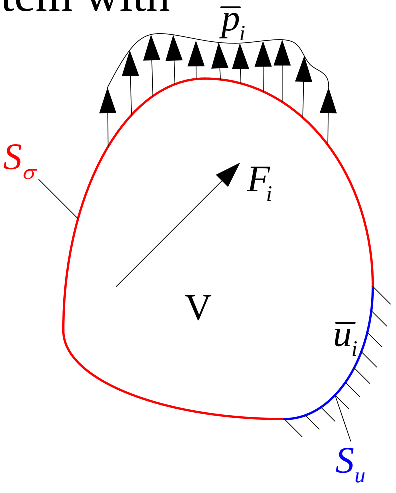
According to the law of conservation of energy, the potential energy will decrease equally when the force system does work to the external.

This is the physical meaning of the equation A = -V

In elastic mechanics the load size and direction are usually irrelevant to the deformation of the body. So it can be regarded as a constant force system with potential energy.

So the potential energy of the external force in the figure on the right can be written as

$$V = -\int_V F_i u_i dV - \int_{S_\sigma} \bar{p}_i u_i dS$$



So the total potential energy of the system is

$$\Pi = \int_V W dV - \int_V F_i u_i dV - \int_{S_\sigma} \bar{p}_i u_i dS$$

, where the first term on the right is the strain energy.

5.2 The strain and complementary energies

- **The total potential complementary energy Π^* of the elastic system**

Π^* is defined as the sum of the strain complementary energy U^* and the total potential complementary energy V^* of the supporting system

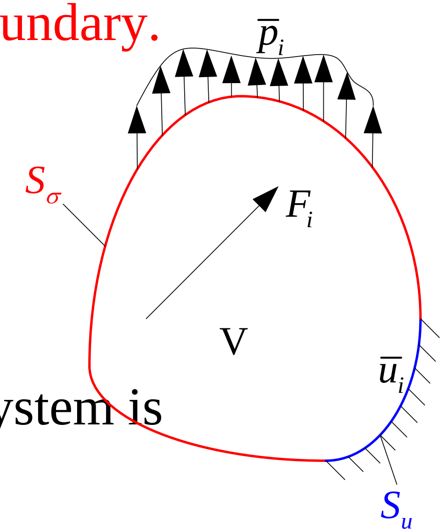
$$\Pi^* = U^* + V^*$$

When the displacement is allowed on the supporting boundary of the elastic system, the surplus energy absorbed by the supporting system or passed to other objects by the supporting system is called the potential complementary energy of the supporting system, denoted as V^*, which equals to the negative value of the work of the reaction force on the given displacement of the boundary.

The reaction force on the displacement boundary S_u is p_i

Then the potential complementary energy of the supporting system is

$$V^* = -\int_{S_u} p_i \bar{u}_i dS$$



Thus the total potential complementary energy of the elastic system is

.

$$\Pi^* = \int_V W^* dV - \int_{S_u} p_i \bar{u}_i dS$$

5.3 The principle of virtual work

❖ The principle of virtual work

- We have learnt the principle of virtual work of the rigid body in theoretical mechanics.

  If an object is in equilibrium in a certain force system, then the total virtual work of the external forces equals zero when the displacement from the equilibrium position is small enough and allowed by the constraint condition (the virtual displacement or the displacement variation).

  This theorem is also the necessary and sufficient condition for the force system to be in equilibrium.

- The theorem before can be extended to the deformable body. What's more we have already learnt to use the principle of virtual work of the line system in the material mechanics and the structural mechanics.

  The related basic concepts can be derived similarly, which can be found in the textbooks. Here we recommend "the Structural Mechanics" edited by Long Yuqiu.

- Now we'll discuss the principle of virtual work in general elastic mechanics.

5.3 The principle of virtual work

- **The permissible displacement and stress**

  The stress, strain and the displacement are real and unique if exactly satisfy all the equations and boundary conditions.

  It should be noticed that the so-called real state is the exact solution of the corresponding theoretical system.

  For example in some approximate theories like beams, plates and shells, call the exact solutions that satisfy all the conditions as the real state, though the condition is approximate itself.

  So the so-called "real" is mathematically true but not physically true.

  For instance, the stress solutions only satisfying the static equilibrium and the displacements (strain) only satisfying continuous deformation can be many but they are not necessarily the real solutions because maybe they don't satisfy all the conditions.

  The methods we'll discuss here is to find the solutions that satisfy all the remaining conditions amongst the possible solutions. And we call them **real solutions.**

  The remaining conditions (part of differential equations and boundary conditions) can be transformed into the extremum condition of functional.

  Let's first introduce the concept of permissible displacement and stress.

5.3 The principle of virtual work

- **The permissible displacement and stress**
  - **permissible displacement**

    The displacement field that satisfies **the continuous deformation and boundary displacement condition** is called the geometrically permitted displacement.

    The strain that satisfies the **geometric equations or derived from the permissible displacement by the geometric equations** is called the permissible strain.

    Apparently there are an infinite number of permissible displacements and the permissible strain. Only those that both satisfy the static equilibrium are the real displacement and strain.

  - **permissible stress**
  - The stress field that satisfy **the equilibrium equation and the boundary condition** is called the permitted stress of static force.

    The stress field is not unique. The permissible stress that can satisfy the deformation compatibility condition is the real stress.
  - the stress obtained from the permissible strain by the constitutive equations doesn't necessarily satisfy the equilibrium condition, which means it may be not the permissible stress.

    Similarly the strain obtained from the permissible stress by the constitutive equations doesn't necessarily satisfy the deformation compatibility condition. Even one can obtain the corresponding displacement, it may not satisfy the boundary condition, which means it may be not the permissible displacement.

The permissible state provides flexibility to the variation method. During the 200 years development of energy method, there occurred all kinds of solutions. Their main distinction is choosing different independent functions, functionals and the possible states.

5.3 The principle of virtual work

The virtual displacem
actually the variatio
the displacemen

❖ Virtual displacement and stress

The virtual displacement means the infinitely small displacement from one possible position to an adjacent possible position, denoted as δu_i, which is arbitrary.

Of course the virtual displacement satisfies the deformation continuous condition in the domain that is the geometric equation. And we have $\delta u_i=0$ on the boundary.

The corresponding virtual stress also satisfies the geometric equation that is

$$\delta\varepsilon_{ij}=\frac{1}{2}\left(\delta u_{i,j}+\delta u_{j,i}\right)$$

It should be noticed that the permissible displacement is limited small according to the small displacement hypothesis of elastic mechanics. While the virtual displacement is infinitely small.

The virtual displacement and the permissible displacement both satisfy the deformation continuous condition in the domain. On the boundary S_u of the given displacement, the permissible displacement u_i must equal to the given value while δu_i must equals zero.

5.3 The principle of virtual work

❖ Virtual displacement and stress

The virtual stress means the infinitely small variation from one permissible stress to another permissible stress, denoted as $\delta\sigma_{ij}$.

Under the necessary condition that it must equilibrate the given external force, it can be arbitrary.

According to the equilibrium equations and the boundary condition of the given force we have

inside V $$\delta\sigma_{ij,j} = 0 \quad (V内)$$

over Sσ $$\delta\sigma_{ij} l_j = 0 \quad (S_\sigma内)$$

Similar to the virtual displacement, the virtual stress is the variation of the permissible stress so it's infinitely small.

5.3 The principle of virtual work

❖The principle of virtual work

Now let's prove the principle of virtual work that is applicable to all the deformable bodies (small deformation).

We can furtherly derive another two principles, which are the principles of virtual displacement and virtual stress.

➢ The principle of virtual work

For a deformable body in an equilibrium state under a given external force $(F_i, \bar{p}_i)$, the virtual work of the external force on the geometrically permitted displacement u_i equals to the virtual work of the equilibrium permitted stress σ_{ij} on the geometrically permitted strain ε_{ij}. That is

$$\int_V F_i u_i dV + \int_S p_i u_i dS = \int_V \sigma_{ij}\varepsilon_{ij} dV$$

Or decompose the boundary into the following two parts

$$\int_V F_i u_i dV + \int_{S_u} p_i \bar{u}_i dS + \int_{S_\sigma} \bar{p}_i u_i dS = \int_V \sigma_{ij}\varepsilon_{ij} dV$$

➢ F_i, $\bar{p}_i$ is the given body force and the surface force. $\bar{u}_i$ Is the given displacement. V is the occupied space of the elastic body. S is the whole surface of the deformable body. S_u is the surface of the given displacement and S_σ is the surface of the given surface force.

5.3 The principle of virtual work

The principle of virtual work

- The permissible stress and displacement σ_{ij} ,u_i and ε_{ij} should satisfy the following equilibrium conditions and geometric conditions.

$$\left.\begin{aligned} \sigma_{ij,j}+F_i=0 \quad (V内) \\ \sigma_{ij}l_i=\bar{p}_i \quad (S_\sigma上) \end{aligned}\right\} \quad (平衡条件) \qquad \text{Equilibrium condition}$$

$$\left.\begin{aligned} \varepsilon_{ij}=\frac{1}{2}\left(u_{i,j}+u_{j,i}\right) \quad (V内) \\ u_i=\bar{u}_i \quad (S_u上) \end{aligned}\right\} \quad (几何条件) \qquad \text{Geometric condition}$$

The two equations are actually the necessary and sufficient conditions for the principle of virtual work to be valid, which is applicable for every permissible stress and displacement.

- It should be noticed that the permissible stress and permissible displacement (strain) may be not relevant to each other, which means they can vary independently in respective permitted condition.

  The condition doesn't involve the constitutive relationship of the material so it is applicable for any deformable body of small deformation (including linear, non-linear elastic body and the elastic-plastic body).

5.3 The principle of virtual work

- **The principle of virtual work**

  Let's first prove its sufficiency

  - **Prove**

    According to the Cauchy formula of stress and the Gauss integral formula we have

$$\int_S p_i u_i dS = \int_S \sigma_{ij} l_j u_i dS = \int_V (\sigma_{ij} u_i)_{,j} dV$$

$$= \int_V \sigma_{ij,j} u_i dV + \int_V \sigma_{ij} u_{i,j} dV$$

Because of the symmetry of stress tensor and the geometrical condition, the second term above can be written as

$$\int_V \sigma_{ij} u_{i,j} dV = \int_V \sigma_{ij} \frac{1}{2}(u_{i,j} + u_{j,i}) dV = \int_V \sigma_{ij} \varepsilon_{ij} dV$$

  - **So**

$$\int_S p_i u_i dS = \int_V \sigma_{ij,j} u_i dV + \int_V \sigma_{ij} \varepsilon_{ij} dV = \int_{S_u} p_i \bar{u}_i dS + \int_{S_\sigma} \bar{p}_i u_i dS$$

5.3 The principle of virtual work

The principle of virtual work

- **Prove:**

  Substitute the equation of the principle of virtual work and we have

$$\int_V F_i u_i dV + \int_{S_u} p_i \bar{u}_i dS + \int_{S_\sigma} \bar{p}_i u_i dS$$

$$= \int_V F_i u_i dV + \int_V \sigma_{ij,j} u_i dV + \int_V \sigma_{ij}\varepsilon_{ij} dV$$

$$= \int_V (F_i + \sigma_{ij,j}) u_i dV + \int_V \sigma_{ij}\varepsilon_{ij} dV$$

Notice the equilibrium equation thus

$$\int_V F_i u_i dV + \int_{S_u} p_i \bar{u}_i dS + \int_{S_\sigma} \bar{p}_i u_i dS = \int_V \sigma_{ij}\varepsilon_{ij} dV$$

- It shows that the equation always holds when σ_{ij} and u_i (ε_{ij}) satisfy the equilibrium conditions and geometric conditions respectively.

  The inverse proposition of this identical equation also holds, which is actually the principle of virtual displacement and stress. That is to prove the necessity of equilibrium and geometric conditions.

5.3 The principle of virtual work

- **The reciprocal theorem**

As is mentioned before that the principle of virtual work doesn't involve the constitutive relationship of the material, which means it's applicable for any deformable body of small deformation.

It can derive the reciprocal theorem when applied to the linear elastic body.

The principle of virtual work holds for any permissible static stress field and geometric displacement field and holds of course for the real fields. Now let's discuss its application in two different real states of one object.

- Suppose the body force and surface force of the first state are F_i^1, p_i^1, the corresponding stress and deformation are $\sigma_{ij}^1, \varepsilon_{ij}^1, u_i^1$.

The body force and surface force of the second state are F_i^2, p_i^2, the corresponding stress and deformation are $\sigma_{ij}^2, \varepsilon_{ij}^2, u_i^2$.

The two states are both the permissible static states and geometric state because they are real.

Now first let the first state be the permissible static field and the second be the permissible geometric field and then substituted into the principle of virtual work.

$$\int_V F_i^1 u_i^2 dV + \int_S p_i^1 u_i^2 dS = \int_V \sigma_{ij}^1 \varepsilon_{ij}^2 dV$$

Then let the first state be the permissible geometric field and the second be the permissible static field and then substituted into the principle of virtual work.

$$\int_V F_i^2 u_i^1 dV + \int_S p_i^2 u_i^1 dS = \int_V \sigma_{ij}^2 \varepsilon_{ij}^1 dV$$

5.3 The principle of virtual work

- **The reciprocal theorem**

The constitutive relation of the linear elastic material obeys the Hooke's law
$$\sigma_{ij} = C_{ijkl}\varepsilon_{kl}$$

Because of the symmetry of the elastic tensor we have

$$\sigma_{ij}^1\varepsilon_{ij}^2 = C_{ijkl}\varepsilon_{kl}^1\varepsilon_{ij}^2 = C_{klij}\varepsilon_{ij}^2\varepsilon_{kl}^1 = \sigma_{kl}^2\varepsilon_{kl}^1 = \sigma_{ij}^2\varepsilon_{ij}^1$$

- Thus

$$\int_V F_i^1 u_i^2 dV + \int_S p_i^1 u_i^2 dS = \int_V F_i^2 u_i^1 dV + \int_S p_i^2 u_i^1 dS$$

This is the famous reciprocal theorem.

The work of the external force in the first state on the displacement in the second state equals to that of the external force in the second state on the displacement in the first state in an elastic body.

It's also called the Bette formula.

- One can easily obtain the displacement of elastic body using this theorem.

5.3 The principle of virtual work

The reciprocal theorem

- **Example**

  Apply a pair of pressure P along the ends of line L to an arbitrary shaped elastic body. Solve the volume change caused by P of the object.

Solve

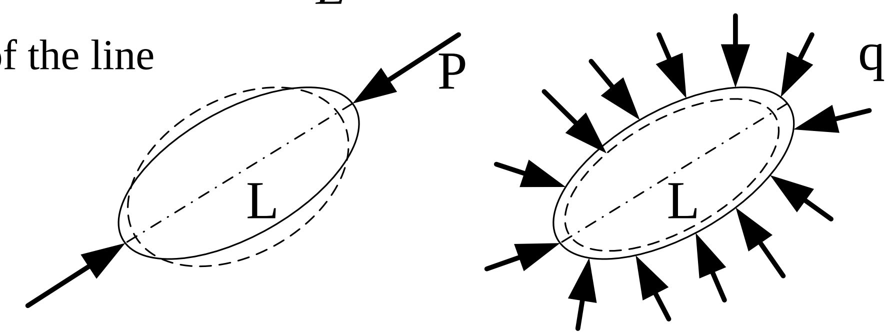
Let the volume change caused by P be ΔV.

Suppose of applying a uniformly distributed unit pressure q to the object and the stress state inside will be the hydrostatic pressure.

So according to the Hooke's Law we have $\varepsilon_x = \varepsilon_y = \varepsilon_z = -\dfrac{q}{E}(1-2\nu)$

Then we can obtain the shortening amount of the line

$$\Delta_L = \varepsilon_L L = -\frac{(1-2\nu)}{E} qL$$



According to the reciprocal theorem

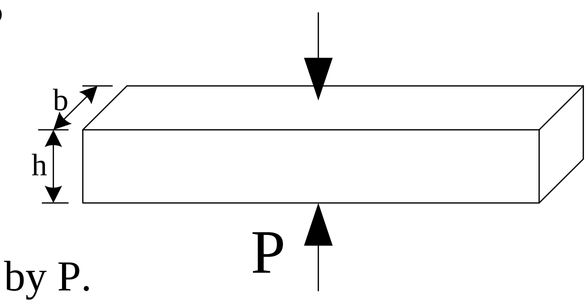
$$P\,\Delta_L = q\,\Delta V$$

Let q = 1 and we can solve the volume change caused by P

$$\Delta V = -\frac{(1-2\nu)}{E} PL$$

- **Exercise**

  Apply a pair of holding force to the middle of the bar with rectangular section. Solve the length variation caused by P.



5.3 The principle of virtual work

The principle of virtual displacement

- **Definition:**

  For a given deformable body in a static system composed of a stress field and given external force, if the work of the static system on all the permissible displacement and the corresponding permissible strain of the deformable body satisfies the principle of virtual work, then the stress field must be the equilibrium permissible stress of the given external force.

- **Prove:**

  For the deformable body with given external force F_i, $\overline{p}_i$, if a stress field can satisfy the virtual work equations for any two infinitely close permissible displacement u_i^1 and u_i^2 .

$$\int_V F_i u_i^1 dV + \int_{S_u} p_i \overline{u}_i^1 dS + \int_{S_\sigma} \overline{p}_i u_i^1 dS = \int_V \sigma_{ij} \varepsilon_{ij}^1 dV$$

$$\int_V F_i u_i^2 dV + \int_{S_u} p_i \overline{u}_i^2 dS + \int_{S_\sigma} \overline{p}_i u_i^2 dS = \int_V \sigma_{ij} \varepsilon_{ij}^2 dV$$

Subtract one equation from the other and we have

5.3 The principle of virtual work

The principle of virtual displacement

- **Prove:**

$$\int_V F_i(u_i^1-u_i^2)dV+\int_{S_u} p_i(\bar{u}_i^1-\bar{u}_i^2)dV+\int_{S_\sigma}\bar{p}_i(u_i^1-u_i^2)dV=\int_V \sigma_{ij}(\varepsilon_{ij}^1-\varepsilon_{ij}^2)dV$$

Σ_{ij} is given and the u_i is given on S_u. Ignore the second term.

$$\int_V F_i(u_i^1-u_i^2)dV+\int_{S_\sigma}\bar{p}_i(u_i^1-u_i^2)dV=\int_V \sigma_{ij}(\varepsilon_{ij}^1-\varepsilon_{ij}^2)dV$$

Let

$$\delta u_i=u_i^1-u_i^2$$

$$\delta\varepsilon_{ij}=\varepsilon_{ij}^1-\varepsilon_{ij}^2=\frac{1}{2}(u_{i,j}^1+u_{j,i}^1)-\frac{1}{2}(u_{i,j}^2+u_{j,i}^2)=\frac{1}{2}(\delta u_{i,j}+\delta u_{j,i})$$

The equation can be simplified as

$$\int_V F_i\delta u_i\,dV+\int_{S_\sigma}\bar{p}_i\delta u_i\,dS=\int_V \sigma_{ij}\delta\varepsilon_{ij}dV$$

Δu_i is apparently the virtual displacement, which can be regarded as the variation of displacement.

5.3 The principle of virtual work

- **The principle of virtual displacement**
  - **Prove:**

We can have $\int_V F_i \delta u_i \, dV + \int_{S_\sigma} \bar{p}_i \delta u_i \, dS = \int_V \sigma_{ij} \delta \varepsilon_{ij} dV$

Integrate the right part of the equation and transform as follows

$$\int_V \sigma_{ij} \delta \varepsilon_{ij} dV = \int_V \sigma_{ij} \frac{1}{2}(\delta u_{i,j} + \delta u_{j,i}) dV = \int_V \frac{1}{2}(\sigma_{ij} \delta u_{i,j} + \sigma_{ji} \delta u_{j,i}) dV$$

$$= \int_V \sigma_{ij} \delta u_{i,j} dV = \int_V (\sigma_{ij} \delta u_i)_{,j} dV - \int_V \sigma_{ij,j} \delta u_i \, dV = \int_S \sigma_{ij} \delta u_i l_j dS - \int_V \sigma_{ij,j} \delta u_i \, dV$$

For the partial boundary S_u of the boundary S with given displacement, we have $\delta u_i = 0$, thus

$$\int_V \sigma_{ij} \delta \varepsilon_{ij} dV = \int_{S_\sigma} \sigma_{ij} \delta u_i l_j dS - \int_V \sigma_{ij,j} \delta u_i \, dV$$

Substituted into the first equation and we have

$$\int_V (F_i + \sigma_{ij,j}) \delta u_i \, dV + \int_{S_\sigma} (\bar{p}_i - \sigma_{ij} l_j) \delta u_i \, dS = 0$$

The integrand function equals zero according to the preparation theorem of variation and we can prove the stress is equilibrium permitted.

5.3 The principle of virtual work

The principle of virtual stress

- **Definition:**

  For a deformable body with a given external force in a displacement field and the corresponding strain field, if both the given force and any equilibrium permittable stress can satisfy the equation of virtual work, then this displacement field is geometrically permitted.

- **Prove:**

  For the given external force F_i, $\overline{p}_i$, if the permissible stress σ_{ij}^1 and σ_{ij}^2 of any two arbitrary adjacent states can satisfy the equation of virtual work in a certain displacement field u_i and the according strain field ε_{ij}

$$\int_V F_i u_i \, dV + \int_{S_u} p_i^2 \overline{u}_i dS + \int_{S_\sigma} \overline{p}_i \, u_i dS = \int_V \sigma_{ij}^2 \varepsilon_{ij} dV$$

$$\int_V F_i u_i \, dV + \int_{S_u} p_i^2 \overline{u}_i dS + \int_{S_\sigma} \overline{p}_i \, u_i dS = \int_V \sigma_{ij}^1 \varepsilon_{ij} dV$$

Because F_i, $\overline{p}_i$ are given. Subtract one equation from the other. Thus

$$\int_{S_u} \delta p_i \overline{u}_i dS = \int_V \delta\sigma_{ij} \varepsilon_{ij} dV$$

Let $\delta F = 0, \delta \overline{p}_i = 0$ and the equation can be equally written as

5.3 The principle of virtual work

- **The principle of virtual stress**
  - **Prove:**

$$\int_V \delta F_i u_i \, dV + \int_{S_u} \delta p_i \bar{u}_i dS + \int_{S_\sigma} \delta \bar{p}_i u_i dS = \int_V \delta\sigma_{ij}\varepsilon_{ij} dV$$

Where

$$\delta\sigma_{ij} = \sigma_{ij}^1 - \sigma_{ij}^2$$

$$\delta p_i = p_i^1 - p_i^2$$

They are arbitrary infinitesimal. $\delta\sigma_{ij}$ is the virtual stress that can be regarded as the variation of stress here. Δp_i is the variation of the surface force (on the boundary S_σ of the given surface force we have $\delta p_i = 0$).

According to the definition of the equilibrium permitted permissible stress we have

(in V) $$\delta F_i = 0 = -\delta\sigma_{ij,j} \quad (V内)$$

(on S_σ) $$\delta \bar{p}_i = 0 = \delta\sigma_{ij} l_j \quad (S_\sigma上)$$

Substituted into the left part of the first equation then

5.3 The principle of virtual work

- **The principle of virtual stress**
  - **Prove:**

$$\int_V -\delta\sigma_{ij,j} u_i \, dV + \int_{S_u} \delta p_i \bar{u}_i dS + \int_{S_\sigma} \delta\sigma_{ij} l_j u_i dS$$

Transform the first term

$$-\int_V \delta\sigma_{ij} u_i \, dV = -\int_V (\delta\sigma_{ij} u_i)_{,j} dV + \int_V \delta\sigma_{ij} u_{i,j} dV$$

$$= -\int_S \delta\sigma_{ij} u_i l_j dS + \int_V \delta\sigma_{ij,j} \frac{1}{2}(u_{i,j} + u_{j,i}) dV$$

Substituted into the equation above and eliminate the integral term on the boundary S_σ of the given external force and we have

$$\int_{S_u} \delta p_i (\bar{u}_i - u_i) dS + \int_V \delta\sigma_{ij} \frac{1}{2}(u_{i,j} + u_{j,i}) dV$$

Substituted into the first equation on the previous slide.

$$\int_V \delta\sigma_{ij} [\varepsilon_{ij} - \frac{1}{2}(u_{i,j} + u_{j,i})] dV + \int_S \delta p_i (u_i - \bar{u}_i) dS = 0$$

The integrand function equals zero according to the preliminary theorem of variation and we can prove the displacement is geometrically permitted.

5.3 The principle of virtual work

Conclusion

- **We can conclude that**

  The principle of virtual displacement is equivalent to the equilibrium equation and the boundary condition of external force.

  The principle of virtual stress is equivalent to the geometric equation and the boundary condition of displacement.

- In terms of functional and variation

  The independent function of the principle of virtual displacement is the permissible displacement u_i, and its independent variation is δu_i 。

  The independent function of the principle of virtual stress is the permissible stress σ_{ij}, and its independent variation is $\delta\sigma_{ij}$.

  It shows that the boundary problems of the differential equations of elastic mechanics can be expressed in the form of variation equations. The variation equation of the equilibrium equation is the principle of virtual displacement. The variation equation of the geometric equation is the principle of virtual stress.

- The principles of virtual displacement and stress don't involve the constitutive relationship of the material so they are applicable for any deformable body of small deformation.

  The next two chapters will show how to derive the two classical variation principles "the principle of minimum potential energy and minimum complementary energy" from the principle of virtual displacement and stress.

5.4 The Principle of Minimum Potential Energy

According to the Green theorem we have the function W of the strain for an ideal elastic body

$$\sigma_{ij} = \frac{\partial W}{\partial \varepsilon_{ij}}$$

Now suppose that the stress and displacement of the principle of virtual displacement are not independent but dependent on each other.

So we can have the equation of virtual displacement according to the Green theorem that

$$\int_V F_i \delta u_i \, dV + \int_{S_\sigma} \overline{p}_i \delta u_i \, dS = \int_V \frac{\partial W}{\partial \varepsilon_{ij}} \delta \varepsilon_{ij} dV$$

This is the result of applying the principle of virtual displacement to the ideal elastic body.

The equation is equivalent to the equilibrium equation according to the principle of virtual displacement. But it should be noticed that the strain here is not the independent variation but expressed in the form of displacement according to the geometric equations.

Meanwhile the specific constitute relation is introduced here so the equations derived above is the equilibrium equations expressed by displacement.

Exercise：

Derive the equilibrium equations expressed by displacement by the principle of virtual displacement for the isotropy linear elastic body.

Hints：

Replace $\frac{\partial W}{\partial \varepsilon_{ij}}$ in the equation above by the Hooke's law (adopt the expression of Lame constants)

5.4 The Principle of Minimum Potential Energy

In the principle of virtual displacement, u_i has already satisfied the geometric equations and the boundary condition of displacement in advance.

Now it also satisfies the constitutive relationship, which expresses the equilibrium condition equivalent to the principle of virtual displacement. So the permissible displacement mentioned above can satisfy both the equilibrium conditions and the boundary conditions.

So the displacement must be the real one and we can conclude that the geometric permissible displacement satisfying the following equation must be the real. ：

$$\int_V F_i \delta u_i \, dV + \int_{S_\sigma} \bar{p}_i \delta u_i \, dS = \int_V \frac{\partial W}{\partial \varepsilon_{ij}} \delta \varepsilon_{ij} dV$$

The total potential energy Π of the elastic body consists of the strain energy U and the potential energy V of the external force. That is

$$\Pi = U + V$$

For the potential energy of the external force

$$\delta V = \delta(-\int_V F_i u_i dV - \int_{S_\sigma} \bar{p}_i u_i dS) = -(\int_V F_i \delta u_i dV + \int_{S_\sigma} \bar{p}_i \delta u_i dS)$$

For the strain energy

$$\delta U = \delta \int_V W dV = \int_V \delta W dV = \int_V \frac{\partial W}{\partial \varepsilon_{ij}} \delta \varepsilon_{ij} dV$$

According to the principle of virtual displacement we can conclude form the equations above that $\delta\Pi = \delta(U+V) = \delta U + \delta V = 0$

5.4 The Principle of Minimum Potential Energy

From the discussion of the functional extremum we have

$$\delta\Pi = \delta(U+V) = \delta U+\delta V=0$$

which is the condition that the total potential energy of the elastic system can reach its extremum.

This result is obtained directly by the principle of virtual displacement when the given external force is conservative (so there exists the potential energy of external force).

So the geometric permissible displacement that satisfy the equation above must be the **real one.**

In another word the permissible displacement that make the total energy of the elastic system get the extremum must be the real displacement, vice versa.

It can be further proven that the second-order variation of the total potential energy of the elastic system for the real displacement is

$$\delta^2\Pi \geqq 0$$

So the **principle of minimum potential energy** of elastic mechanics is

For the elastic system of small deformation under the action of given external force, the real displacement field satisfying the equilibrium condition among all the possible ones can get the minimum of potential energy.

Or:

The permissible displacement field that make the elastic system reach its minimum potential energy is the real one.

The fundamental idea of the finite element method is to obtain the real displacement among all the possible ones by means of **the principle of minimum potential energy**. And we can obtain the principle of minimum complementary energy in the same way.

5.5 The Euler equations of elastic mechanics

- The principles of minimum potential energy and complementary energy are two fundamental variation principles with two kinds of solutions: the Euler method and direct method.

- The Euler method is to derive the corresponding differential equations and boundary conditions on the basis of variation theorem to establish the equivalent relation between the variation and differential method of elastic mechanics.
  - The Euler equation derived from the principle of minimum potential energy is the **equilibrium equations** of elastic mechanics, which is the criterion to distinguish the real state and the static possible state.
  - The Euler equation derived from the principle of minimum complementary energy is the **compatibility equation (geometric equations)**, which is the criterion to distinguish the real state and geometrically possible state.
- The process of deriving the differential equations of elastic mechanics problems by the variation theorem is very strict and one can also derive the corresponding boundary condition at the same time.

  These systematic deduction methods help prove the rationale of the the equivalent force boundary condition in the theory of plates and shells.

  The equivalent force boundary condition is to transform the torque on the boundary into the equivalent shear and tensile forces.

5.5 The Euler equations of elastic mechanics

- **Example: the plane stress problem**
  We'll derive the plane stress problem here for example. Because the question is quite simple, we don't adopt the tensor form.

Solve:

The expression of strain energy of elastic mechanics problem.

$$U = \frac{1}{2}\iint_{\Omega}(\sigma_x\varepsilon_x + \sigma_y\varepsilon_y + \tau_{xy}\gamma_{xy})dxdy$$

$$= \frac{G}{1-\mu}\iint_{\Omega}(\varepsilon_x^2 + \varepsilon_y^2 + 2\mu\varepsilon_x\varepsilon_y + \frac{1-\nu}{2}\gamma_{xy}^2)dxdy$$

$$= \frac{E}{2(1-\mu^2)}\iint_{\Omega}[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + 2\mu\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} + \frac{1-\nu}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)^2]dxdy$$

The expression of the total potential energy is

$$\Pi = U - \iint_{\Omega}(F_x u + F_y v)dxdy - \int_{\Gamma_\sigma}(\overline{p}_x u + \overline{p}_y v)ds$$

Fx and Fy are the body forces in domain, $\overline{p}_x$, $\overline{p}_y$ are the given surface forces on the boundary. The independent function of the functional is the displacement components u and v.

5.5 The Euler equations of elastic mechanics

According to the principle of minimum potential energy, let

$$\delta\Pi = 0。$$

Do the variation calculation of the potential energy of external force

$$\delta V = -\iint_{\Omega}(F_x\delta u + F_y\delta v)dxdy - \int_{\Gamma_\sigma}(\overline{p}_x\delta u + \overline{p}_y\delta v)ds$$

Then do the variation calculation of the strain energy

$$\delta U = \frac{E}{2(1-\mu^2)}\delta\iint_{\Omega}[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + 2\mu\frac{\partial u}{\partial x}\frac{\partial v}{\partial y} + \frac{1-\mu}{2}\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)^2]dxdy$$

$$= \frac{E}{2(1-\mu^2)}\iint_{\Omega}\left\{2\left(\frac{\partial u}{\partial x} + \mu\frac{\partial v}{\partial y}\right)\delta\left(\frac{\partial u}{\partial x}\right) + 2\left(\frac{\partial v}{\partial y} + \mu\frac{\partial u}{\partial x}\right)\delta\left(\frac{\partial v}{\partial y}\right) + (1-\mu)\left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)\left[\delta\left(\frac{\partial u}{\partial y}\right) + \delta\left(\frac{\partial v}{\partial x}\right)\right]\right\}dxdy$$

According to the integration by parts, we have

$$\iint_{\Omega}2\left(\frac{\partial u}{\partial x} + \mu\frac{\partial v}{\partial y}\right)\delta\left(\frac{\partial u}{\partial x}\right)dxdy = 2\iint_{\Omega}\left(\frac{\partial u}{\partial x} + \mu\frac{\partial v}{\partial y}\right)\frac{\partial}{\partial x}(\delta u)dxdy$$

$$= 2\left\{\iint_{\Omega}\frac{\partial}{\partial x}\left[\delta u\left(\frac{\partial u}{\partial x} + \mu\frac{\partial v}{\partial y}\right)\right]dxdy - \iint_{\Omega}\delta u\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x} + \mu\frac{\partial v}{\partial y}\right)dxdy\right\}$$

Thus

5.5 The Euler equations of differential problems of elastic mechanics

$$\delta U=\frac{E}{(1-\mu^2)}\iint_\Omega \frac{\partial}{\partial x}\left[\delta u\left(\frac{\partial u}{\partial x}+\mu\frac{\partial v}{\partial y}\right)+\frac{1-\mu}{2}\delta v\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]dxdy$$

$$+\frac{E}{(1-\mu^2)}\iint_\Omega \frac{\partial}{\partial x}\left[\delta v\left(\frac{\partial v}{\partial y}+\mu\frac{\partial u}{\partial x}\right)+\frac{1-\mu}{2}\delta u\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]dxdy$$

$$-\frac{E}{(1-\mu^2)}\iint_\Omega \delta u\left[\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial x}+\mu\frac{\partial v}{\partial y}\right)+\frac{1-\mu}{2}\frac{\partial}{\partial y}\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]dxdy$$

$$-\frac{E}{(1-\mu^2)}\iint_\Omega \delta v\left[\frac{\partial}{\partial y}\left(\frac{\partial v}{\partial y}+\mu\frac{\partial u}{\partial x}\right)+\frac{1-\mu}{2}\frac{\partial}{\partial x}\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]dxdy$$

- According to the Green integral formula and including the potential energy due to the external forces

$$\delta\Pi=-\frac{E}{(1-\mu^2)}\iint_\Omega \delta u\left[\left(\frac{\partial^2 u}{\partial x^2}+\mu\frac{\partial^2 v}{\partial x\partial y}\right)+\frac{1-\mu}{2}\left(\frac{\partial^2 u}{\partial^2 y}+\frac{\partial^2 v}{\partial x\partial y}\right)+F_x\right]dxdy$$

$$-\frac{E}{(1-\mu^2)}\iint_\Omega \delta v\left[\left(\frac{\partial^2 v}{\partial y^2}+\mu\frac{\partial^2 u}{\partial x\partial y}\right)+\frac{1-\mu}{2}\left(\frac{\partial^2 v}{\partial^2 y}+\frac{\partial^2 u}{\partial x\partial y}\right)+F_y\right]dxdy$$

$$+\frac{E}{(1-\mu^2)}\int_\Gamma \delta u\left[l\left(\frac{\partial u}{\partial x}+\mu\frac{\partial v}{\partial y}\right)+m\frac{1-\mu}{2}\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]ds+\frac{E}{(1-\mu^2)}\int_\Gamma \delta v\left[m\left(\frac{\partial v}{\partial y}+\mu\frac{\partial u}{\partial x}\right)+l\frac{1-\mu}{2}\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]ds$$

$$-\int_{\Gamma_\sigma}(\bar{p}_x\delta u+\bar{p}_y\delta v)ds$$

5.5 The Euler equations of differential problems of elastic mechanics

- According to the preliminary theorem of variation, we have

(in Ω)

$$\frac{E}{(1-\mu^2)}\left(\frac{\partial^2 u}{\partial x^2}++\frac{1-\mu}{2}\frac{\partial^2 u}{\partial^2 y}+\frac{1+\mu}{2}\frac{\partial^2 v}{\partial x\partial y}\right)+F_x=0$$

$$\frac{E}{(1-\mu^2)}\left(\frac{\partial^2 v}{\partial y^2}++\frac{1-\mu}{2}\frac{\partial^2 v}{\partial x^2}+\frac{1+\mu}{2}\frac{\partial^2 u}{\partial x\partial y}\right)+F_x=0$$

在Ω内

- Divide the boundary Γ into two parts, and one can obtain the natural boundary condition on Γ_σ.

(on Γ_σ)

$$\frac{E}{(1-\mu^2)}\left[l\left(\frac{\partial u}{\partial x}+\mu\frac{\partial v}{\partial y}\right)+m\frac{1-\mu}{2}\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]=\bar{p}_x$$

$$\frac{E}{(1-\mu^2)}\left[m\left(\frac{\partial v}{\partial y}+\mu\frac{\partial u}{\partial x}\right)+l\frac{1-\mu}{2}\left(\frac{\partial u}{\partial y}+\frac{\partial v}{\partial x}\right)\right]=\bar{p}_y$$

在Γ_σ上

- The essential boundary condition on Γ_u is

(on Γu)

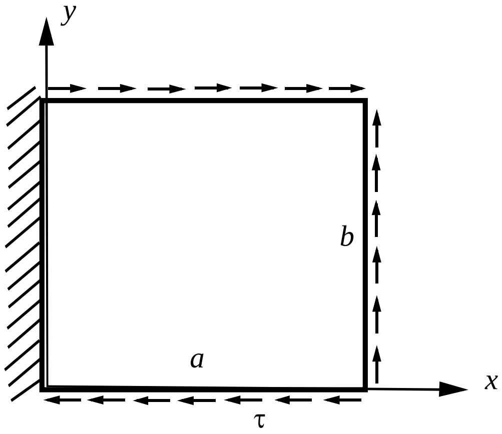
$$u=\bar{u},\quad v=\bar{v}\quad 在\Gamma_u上$$

5.6 Examples

5-6-1 A thin plate has one fixed edge and three free edges. The shearing stress τ applied to the free edges is uniformly distributed. Ignore the body force. Let the displacement component

$$u = x(A_1 + A_2 x + A_3 y + \ldots) \qquad v = x(B_1 + B_2 x + B_3 y + \ldots)$$

Try to solve the displacement of the thin plate by Ritz method.



5.6 Examples

The displacement satisfies the boundary condition of the fixed edge, which means $(u)_{x=0}=0$, $(v)_{x=0}=0$, so it's the permissible displacement.
The strain energy is

$$U=\iint \frac{1}{2}\left(\sigma_x \varepsilon_x+\sigma_y \varepsilon_y+\tau_{xy} \gamma_{xy}\right) dxdy$$

Expressed by displacement

$$U=\frac{E}{2\left(1-\nu^2\right)} \int_0^a \int_0^b\left[\left(\frac{\partial u}{\partial x}\right)^2+\left(\frac{\partial v}{\partial y}\right)^2+2 \nu \frac{\partial u}{\partial x} \frac{\partial v}{\partial y}+\frac{1-\nu}{2}\left(\frac{\partial v}{\partial x}+\frac{\partial u}{\partial y}\right)^2\right] dxdy$$

Take only one term as the approximate solution of displacement.

$$U=\frac{Eab}{2\left(1-\nu^2\right)}\left(A_1^2+\frac{1-\nu}{2} B_1^2\right)$$

5.6 Examples

When $y=0$, $\overline{X} = -\tau$, $\overline{Y} = 0$,
When $y=b$, $\overline{X} = \tau$, $\overline{Y} = 0$,
When $x=a$, $\overline{X} = 0$, $\overline{Y} = \tau$,

The potential energy of external force is

$$V = -\int_{\Gamma} \left(\overline{X} u + \overline{Y} v + \overline{W} w \right) d\Gamma = -\left[\int_0^{q} (-\tau) A_1 x dx + \int_0^{b} \tau B_1 a dy + \int_0^{a} \tau A_1 x dx \right] = -\tau B_1 ab$$

The total potential energy is

$$\Pi = U + V = \frac{Eab}{2\left(1-\nu^2\right)} \left(A_1^2 + \frac{1-\nu}{2} B_1^2 \right) - \tau B_1 ab$$

$$\frac{\partial \Pi}{\partial A_1} = 0 \qquad \frac{\partial \Pi}{\partial B_1} = 0$$

Thus

$$A_1 = 0, \quad B_1 = \frac{2(1+\nu)}{E} \tau$$

5.6 Examples

5-6-2: Solve the deflection of a simply supported beam under uniformly distributed load by Galerkin method.

• The displacement function must satisfy **both the displacement boundary condition and the force boundary condition**.

The displacement boundary condition is

$$w|_{x=0,\ x=l} = 0$$

The bending moment equals zero on the force boundary.

$$w''|_{x=0,\ x=l} = 0$$

Choose the displacement function in form of polynomial, which is

$$w=a_1x^4+a_2x^3+a_3x^2+a_4x+a_5$$

If the displacement function satisfy both the boundary conditions of displacement and force, then

$$a_3=a_5=0 \qquad a_2=-2a_1l \quad a_4=a_1l^3$$

5.6 Examples

- So the displacement function is

$$w=a_1(x^4-2lx^3+l^3x)$$

- The equilibrium equation of the beam is

$$EIw''''-q=0$$

- Use the Galerkin method. Notice $w_1=x^4-2lx^3+l^3x$, so the approximate equation is

$$\int_0^l (EIw''''-q)(x^4-2lx^3+l^3x)dx=0$$

- So

$$a_1=q/24EI.$$

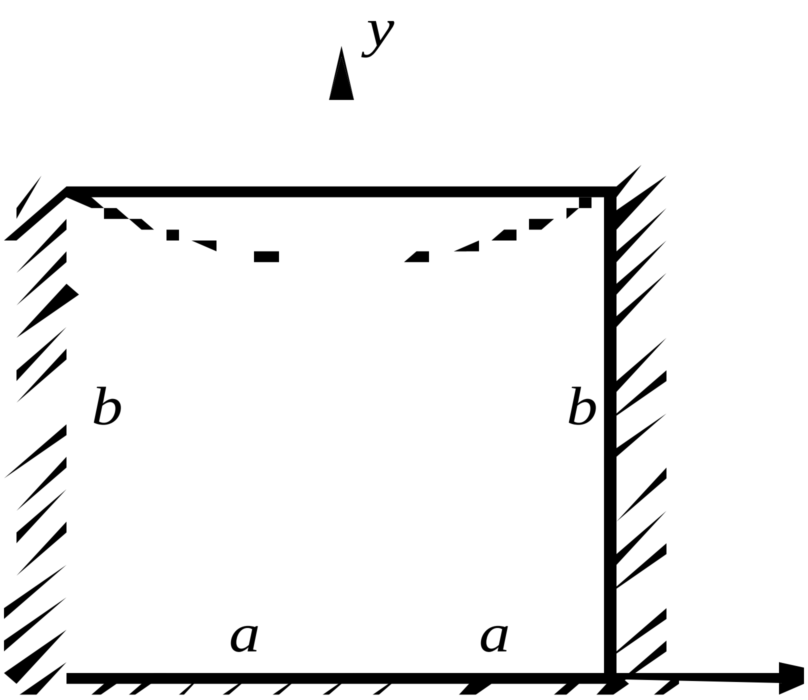
- Thus

$$w=\frac{q}{24EI}(x^4-2lx^3+l^3x)$$

This is the solution of the classic beam theorem.

5.6 Examples

5-6-3 A thin rectangular plate is 2 a in width and b in height. The left, the right and the bottom edges are fixed. And the displacement of the top edge is $u = 0$, $v = -\eta\left(1-\frac{x^2}{a^2}\right)$, ignore the body force and try to solve the displacement and stress of the thin plate.



5.6 Examples

Solve: In the coordinate system shown in the figure, let the displacement components be

$$\boldsymbol{u} = \boldsymbol{A}_1\boldsymbol{u}_1 = \boldsymbol{A}_1\left(1-\frac{\boldsymbol{x}^2}{\boldsymbol{a}^2}\right)\frac{\boldsymbol{x}}{\boldsymbol{a}}\frac{\boldsymbol{y}}{\boldsymbol{b}}\left(1-\frac{\boldsymbol{y}}{\boldsymbol{b}}\right)$$

$$\boldsymbol{v} = \boldsymbol{v}_0 + \boldsymbol{B}_1\boldsymbol{v}_1 = -\eta\left(1-\frac{\boldsymbol{x}^2}{\boldsymbol{a}^2}\right)\frac{\boldsymbol{y}}{\boldsymbol{b}} + \boldsymbol{B}_1\left(1-\frac{\boldsymbol{x}^2}{\boldsymbol{a}^2}\right)\frac{\boldsymbol{y}}{\boldsymbol{b}}\left(1-\frac{\boldsymbol{y}}{\boldsymbol{b}}\right)$$

The displacement function can satisfy all the displacement boundary conditions. There is no force boundary conditions so one can use the Galerkin method directly to simplify the calculation.

Express the equilibrium equation with displacement, so the Glerkin can be shown as

$$\int_{-\boldsymbol{a}}^{\boldsymbol{a}}\int_0^{\boldsymbol{b}}\left(\frac{\partial^2 u}{\partial x^2}+\frac{1-\nu}{2}\frac{\partial^2 u}{\partial y^2}+\frac{1+\nu}{2}\frac{\partial^2 v}{\partial x\partial y}\right)u_1\delta A_1 dxdy$$
$$+\int_{-\boldsymbol{a}}^{\boldsymbol{a}}\int_0^{\boldsymbol{b}}\left(\frac{\partial^2 v}{\partial y^2}+\frac{1-\nu}{2}\frac{\partial^2 v}{\partial x^2}+\frac{1+\nu}{2}\frac{\partial^2 u}{\partial x\partial y}\right)v_1\delta B_1 dxdy = 0$$

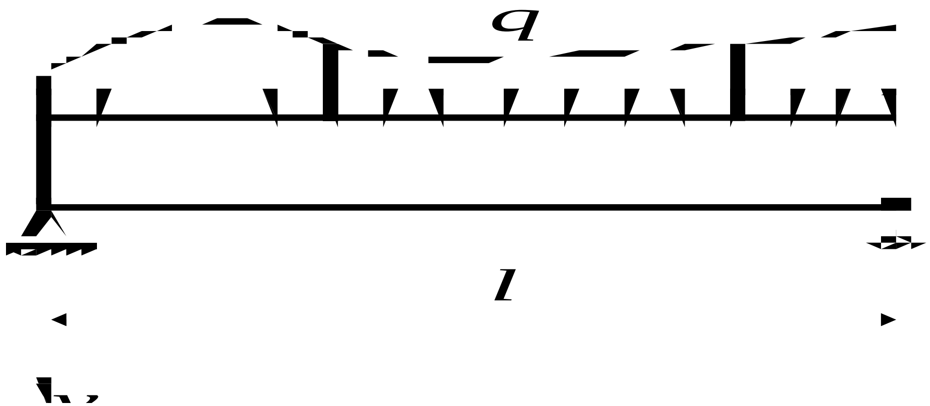
5.6 Examples

Substitute the expression of the approximate displacement and because the variation of the undetermined constants δA_1 and δB_1 are arbitrary, we can obtain two algebraic equations of A_1 and B_1. Thus

$$A_1 = \frac{35(1+\nu)\eta}{42\dfrac{b}{a} + 20(1-\nu)\dfrac{a}{b}} \qquad B_1 = \frac{5(1-\nu)\eta}{16\dfrac{a^2}{b^2} + 2(1-\nu)}$$

5.6 Examples

5-6-4 Apply uniformly distributed load to a simply supported beam. Ignore the body force and derive the equilibrium differential equation and the static boundary condition of the two ends expressed with axis deflection.



Solve：

Denote the deflection as w and ignore the influence of shearing force. Then the strain of the beam can be expressed approximately as

$$U = \frac{1}{2}\int_0^l \frac{M^2}{EI}dx = \frac{1}{2}EI\int_0^l \left(\frac{d^2w}{dx^2}\right)dx$$

5.6 Examples

The potential energy the external load q is $-\int_0^l qwdx$

The total potential energy is

$$\Pi = \frac{EI}{2}\int_0^l \left(\frac{d^2w}{dx^2}\right)^2 dx - \int_0^l qwdx$$

The variation of the total potential energy is

$$\delta\Pi = EI\int_0^l \frac{d^2w}{dx^2}\delta\left(\frac{d^2w}{dx^2}\right)dx - \int_0^l q\delta wdx = 0$$

Calculate the first integral by partial integration

$$= EI\int_0^l \frac{d^2w}{dx^2}\delta\left(\frac{d^2w}{dx^2}\right)dx = EI\int_0^l \frac{d^2w}{dx^2} d\left[\delta\left(\frac{dw}{dx}\right)\right]$$

$$= EI\frac{d^2w}{dx^2}\delta\left(\frac{dw}{dx}\right)\Bigg|_0^l - EI\int_0^l \frac{d^3w}{dx^3}\delta\left(\frac{dw}{dx}\right)dx$$

5.6 Examples

$$= EI\left[\frac{d^2w}{dx^2}\delta\left(\frac{dw}{dx}\right) - \frac{d^3w}{dx^3}\delta w\right]_0^l + EI\int_0^l \frac{d^4w}{dx^4}\delta w dx$$

$$EI\frac{d^4w}{dx^4} - q = 0$$ is the equilibrium equation expressed by deflection

$$EI\left[\frac{d^2w}{dx^2}\delta\left(\frac{dw}{dx}\right) - \frac{d^3w}{dx^3}\delta w\right]_0^l = 0$$

$$\frac{d^2w}{dx^2} = 0$$ The force boundary condition is that the bending moment on the support point equals zero.

5.6 Conclusions

- The Principles of Minimum Potential Energy and Complementary Energy are very fundamental in elastic mechanics, which are often called the classical variation theorems.

- The variation method is to fill all the field equations and boundary conditions of elastic mechanics into two groups expressed in different mathematical forms.
  - One group expressed by the differential field equations and the corresponding boundary conditions. For example the permissible displacement by geometric equations and displacement boundary conditions.
  - The other group expressed by the proper extremum conditions of functional. For example with the principle of minimum potential energy, the equilibrium equation is the one that can make the total potential energy get its minimum.
- The Euler method can solve the problems of elastic mechanics, but it’s not the purpose of introducing variation method.

  Another approach is to solve the variation equations directly, which is called the direct solution of variation problems.

  That is to directly find a set of functions that can make the first-order variation of the functional equal zero.

5.6 Conclusions

- **The direct method of variation problems**

  The basic idea of the direct method is to consider the extremum problems of functional as an extreme condition of the extremum problems of some functions with finite variables or some multivariate functions with finite variables. Let the variables be infinite, then one can find the exact solution.

  In fact it's hard to get infinite variables so the variation theorem is not quite in advantage in solving exact solution, where the direct method is more of the case. It aims to find the optimal solution in a certain subset.

  Of course the selected subset doesn't contain the exact solutions. But the solution will continuously approach to the exact one when enlargeing the subset.

  This is the main idea of the finite element method.

- The functional is naturally established when applying the principles of virtual displacement and stress of the ideal elastic system, so it has a very explicit physical concept.

  It should be noticed that this is one way to establish the functional and there are also other ways such like the trial-and-error method and the method of weighted residuals.

谢谢！